

Combined Vector Field Approach for 2D and 3D Arbitrary Twice Differentiable Curved Path Following with Constrained UAVs

Yueqian Liang · Yingmin Jia

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Abstract Path following is an essential requirement for unmanned aerial vehicles (UAVs). Path following problems of 2-dimensional (2D) straight lines and planar circles have been studied and many approaches have been proposed during the past few years. In this paper the problem of following arbitrary twice differentiable curved paths in both 2D and 3D spaces is investigated. Based on the Helmholtz theorem, respective combined guidance vector fields for any given 2D and 3D arbitrary twice differentiable curved paths are constructed by trading off the respective conservative and solenoidal vector fields making use of the defined distance between the UAV and the desired path. Conditions for the two combined vector fields to be globally feasible are given. UAV input constraints and constant wind disturbance are assumed to be present. By tracking the constructed combined vector fields, a saturated course rate controller for 2D curved path following and a jointly saturated course rate and climb rate controller for 3D curved path following are designed. The Lyapunov stability of the saturated course rate controller for 2D curved path following is

proved. Numerical simulations are given to assess the proposed approach.

Keywords Path following · Unmanned aerial vehicle · Guidance vector field · Input constraint

Mathematics Subject Classification (2010)
93C15 · 93D05 · 93C95 · 34D05

1 Introduction

During the past few decades, unmanned aerial vehicles (UAVs) have found their applications in both military and civilian areas, such as search and rescue, fire detection, traffic monitoring, reconnaissance, surveillance, target protection/tracking/attacking and so on [1–8]. Unlike manned vehicles, to accomplish varieties of missions, unmanned vehicles usually rely on their own autonomy to generate appropriate mission paths and then to follow these paths as accurately as possible [9].

If the desired path is time-related, this corresponds to the trajectory tracking concept. This concept not only requires the vehicle on the path, but also requires it on a particular point of the path at a particular time [10, 11]. Unlike trajectory tracking, if the desired path can be considered to be time-independent, this relates to the path following concept, which only need the vehicle on the path [10, 11]. A typical example of trajectory tracking is to track a maneuvering target. The

Y. Liang (✉) · Y. Jia
The Seventh Research Division and the Center
for Information and Control, Beihang University
(BUAA), Beijing 100191, China
e-mail: liangyueqian3002@126.com

Y. Jia
e-mail: ymjia@buaa.edu.cn

objective is to always keep the target in coverage, so the time requirement is critical. And a typical example of path following is border patrol, the aim of which is to fly along the border and find possible abnormality. Its timeliness is thus not important. Detailed discussions about trajectory tracking and path following can be found in Refs. [10–12] and the references given there.

To address the path following problem, many available approaches have been developed during the past few years (See the recent survey paper [13] and the references therein for more details). Through the Monte-Carlo simulation comparison, Sujit et al. concluded that the guidance vector field approach can make the vehicle follow the desired path more accurately than the other approaches, and also requires the least control effort [13]. This approach was developed by Lawrence [14, 15], Frew [16], Nelson [10], Griffiths [17] et al. The mostly concerned circular path following, which is also referred to as standoff target tracking, was solved by the Lyapunov vector field approach in [16]. A globally convergent Lyapunov guidance vector field was first designed and then by tracking this vector field, the heading rate was commanded to follow the circular path moving with the target [16]. The convoy protection requiring the UAV to fly a racetrack was also discussed by the authors [15, 16]. Vector fields for straight lines and circular paths were constructed in [10], based on which sliding mode controllers were designed to following these two types of paths. In [17], the author further considered the path following of arbitrary curved paths using a modified vector field.

The guidance vector field approach has now been paid much attention by the researchers. Based on the Lyapunov guidance vector field proposed in [16] and by further considering the heading error with respect to the vector field, an alternative feedforward term was exploited to obtain more accurate circular path following in [18, 19]. Recently Zhu et al. gave the rigorous proof of the global convergence of this circular path following approach [20]. By combining Lyapunov guidance vector field and a tangent vector field, Chen et al. further considered this problem temporally to get a theoretically shortest path for circular path following [21]. Bennet et al. studied the autonomous formation reconfiguration problem of a swarm of UAVs through bifurcating a potential vector

field in [22]. In [23], a parameter was introduced to the vector field of [16] to make multiple UAVs simultaneously arrive at their desired positions (which is named capturing mode therein) and achieve their equiangular cooperation. The authors designed a vector field with variable standoff radius and a two-phase approach to achieve the tracking of groups of targets with multiple UAVs in [24]. In [25] the authors investigated the 3-dimensional (3D) straight line following by designing a 3D guidance vector field.

Till now except the work given in [17], the existing results concerning guidance vector field approach for path following only considered paths with particular forms, mostly lines and circles. Although a path following approach for arbitrary 2D paths was proposed in [17], it has four limitations. Firstly, the approach did not consider the UAV input constraint explicitly, which is vital to the UAV flight [26, 27]. Secondly, the designed sliding mode controller is known to suffer the chattering problem due to the controller's inherent discontinuity [28]. Thirdly, the sliding mode controller involves in plenty of parameters and actually four conditions for them (See **Theorem III.1** therein), and it is not an easy work to set appropriate values for these parameters to satisfy these four conditions because some of them depend on a specific curved path (e.g., ρ and $\bar{d}$) and some of them are severely coupled (e.g., μ and $\bar{d}$). Moreover inappropriate parameter settings are apt to lead to large gains and input saturations. And finally, at each sampling instant, a reference point on the curved path should be chosen.

In addition most of the existing results just concentrate on the 2D path following problem. Since in reality the UAVs fly in 3D space, 3D path following should also be an important consideration [25, 29, 30]. As an example, when tracking a target which travels along a 3D trajectory, a UAV must achieve better tracking or sensing performance by changing its own altitude. As another example, when a UAV is flying in a mountainous area, collision avoidance might be more efficient by climbing high than by adjusting the UAV heading. However the efficient guidance vector field approach is rarely applied to the 3D path following in the existing literature.

Motivated by the above discussions, this paper focuses on the UAV path following of arbitrary (twice differentiable) curved paths in both 2D and 3D spaces

by constructing new guidance vector fields. The following aspects are needed to be emphasized.

- The mentioned UAVs throughout this paper are the fixed-wing UAVs rather than the unmanned rotorcrafts. They are assumed to be equipped with low-level flight controllers. The controllers can provide course- and climb-angle-hold functions, and explicit course rate and climb rate can be used as the control inputs. Their airspeeds are assumed to be constant from the fuel efficiency and mission duration points of view.
- Steady background wind disturbance is assumed to be present during the whole path following mission. And we do not consider the wind disturbance in the vertical direction. As usual the constant wind speed is assumed to be less than the constant UAV airspeed.
- The considered curves are assumed to be defined in inertial frame.
- Denote the planar coordinates and the 3D coordinates as $(x, y)^T$ and $(x, y, z)^T$ respectively. Assume that a 2D curve can be represented by an equation $f(x, y) = 0$ and a 3D curve can be defined by a couple of equations $\{f(x, y, z) = 0, g(x, y, z) = 0\}$.
- We say that a 2D curve is twice differentiable if all the first- and second-order partial derivatives of $f(x, y)$ with respect to x and y exist. That is, all of f_x, f_y, f_{xx}, f_{xy} and f_{yy} exist. A similar convention is given for a 3D twice differentiable curve.
- We use the abbreviations “2D curved path following” and “3D curved path following” to represent the phrases “2D arbitrary twice differentiable curved path following” and “3D arbitrary twice differentiable curved path following” respectively.

The contributions of this paper are listed as follows.

- For the 2D curved path following, we construct a new combined vector field by trading off a conservative vector field and a solenoidal vector field based on the Helmholtz theorem.
- For the 2D curved path following, we give a condition for the 2D combined vector field to be globally feasible.
- For the 2D curved path following, we design a saturated course rate controller by explicitly

considering the UAV input constraints, and systematically prove its Lyapunov stability.

- For the 3D curved path following, we construct a combined vector field by trading off a conservative vector field and a solenoidal vector field based on the Helmholtz theorem.
- For the 3D curved path following, we give a condition for the 3D combined vector field to be globally feasible.
- For the 3D curved path following, we design a jointly saturated course rate and climb rate controller by explicitly considering the UAV input constraints.

The remainder of the paper is structured as follows. The arbitrary twice differentiable curved path following in 2D and 3D spaces are studied in Section 2 and Section 3 respectively. The problems are firstly formulated, then the combined vector fields are proposed, and finally the saturated controllers are designed. Section 4 gives some discussions of the proposed approaches. In Section 5 several simulations are conducted to assess the proposed approach. A short conclusion and some future works are given in the last section.

2 2D Curved Path Following

2.1 Problem Formulation

We firstly investigate the 2D path following by holding the altitude constant through zero climb rate. The kinematics model of the UAV in inertial XY frame is given by

$$\dot{x} = V_a \cos \psi + W_x \quad (1)$$

$$\dot{y} = V_a \sin \psi + W_y \quad (2)$$

$$\dot{\psi} = u_{10} \quad (3)$$

where $\mathbf{p} = (x, y)^T$ is the horizontal position of the UAV in inertial frame, $\mathbf{v} = (\dot{x}, \dot{y})^T$ denotes the horizontal inertial velocity, $\psi \in (-\pi, \pi]$ is the UAV heading, V_a is the constant airspeed, and $\mathbf{W} = (W_x, W_y)^T$ is the wind velocity. u_{10} represents the commanded heading rate, satisfying the constraint

$$|u_{10}| \leq \dot{\psi}_{\max} \quad (4)$$

Denoting the magnitude of $\mathbf{v}$ as V_g , it is actually the groundspeed. Note that V_g is variable even for con-

stant airspeed V_a and constant wind speed W . Then by introducing the course χ , the “heading” of the UAV’s ground velocity, Eqs. 1–3 can be equivalently expressed as

$$\dot{x} = V_g \cos \chi \quad (5)$$

$$\dot{y} = V_g \sin \chi \quad (6)$$

$$\dot{\chi} = u_1 \quad (7)$$

where u_1 is the course rate input. The relation between the heading rate input and the course rate input is [16]

$$u_1 = L(\psi)u_{10} \quad (8)$$

where

$$L(\psi) = \frac{V_a^2 + V_a (W_x \cos \psi + W_y \sin \psi)}{V_a^2 + W_x^2 + W_y^2 + 2V_a (W_x \cos \psi + W_y \sin \psi)} \quad (9)$$

Since $\sqrt{W_x^2 + W_y^2} = W < V_a$, we can see that $L(\psi) \in (1/2, 1)$. From Eq. 9 we can see that there also exists a constraint on the course rate, which is assumed to be

$$|u_1| \leq \dot{\chi}_{\max} \quad (10)$$

As an example, $\dot{\chi}_{\max}$ can be set as $\dot{\psi}_{\max}/2$. In addition it can be found that the relation between the airspeed and the groundspeed is

$$V_g = (W_x \cos \chi + W_y \sin \chi) + \sqrt{V_a^2 - (W_x \sin \chi - W_y \cos \chi)^2} \quad (11)$$

Given a desired 2D arbitrary twice differentiable curved path $\mathcal{C}_2 : f(x, y) = 0$, we define a directional distance between the current UAV position $\mathbf{p}$ to $\mathcal{C}_2$ as $r = f(\mathbf{p}) = f(x, y)$. Large $|r|$ implies large distance between them, and $r = 0$ means that $\mathbf{p}$ locates on the desired path.

The 2D arbitrary twice differentiable curved path following problem can now be summarized as: To design input commands u_1 for a fixed-wing UAV whose kinematics is given by Eqs. 5–7 such that (i) the input constraints (10) holds, and (ii) if possible, make the UAV fly along the desired path, and if not, make the absolute distance $|r|$ as small as we can.

2.2 Conservative Vector Field and Solenoidal Vector Field

The Helmholtz theorem states that an arbitrary vector field can be decomposed into two parts, conservative part and solenoidal part. The conservative part is irro-

tational, which can usually be given as the gradient of a scalar function. And the solenoidal part is a rotational one, which can be expressed by the curl of a vector function.

Considering a simple scalar potential function $f_c = r^2 = f^2(x, y)$, its value minimizes at $r = 0$, which

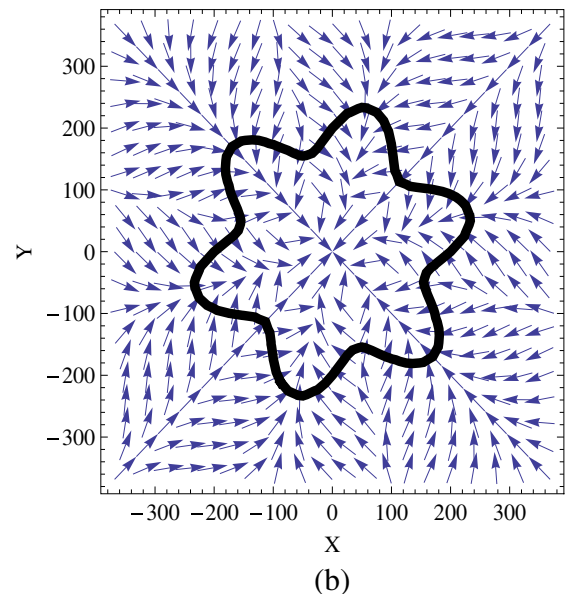
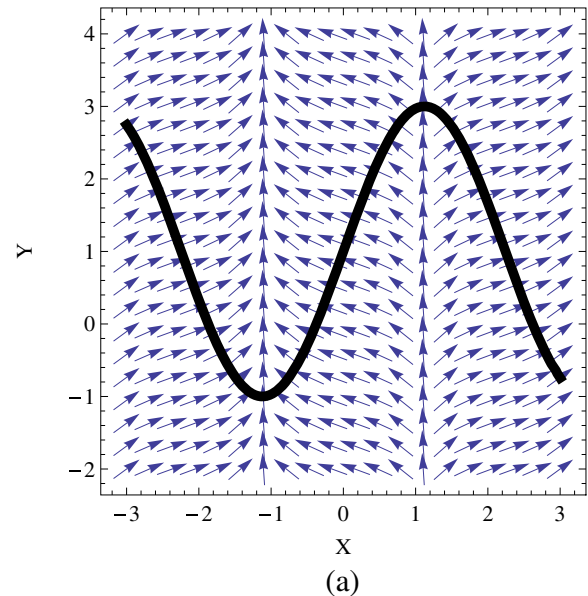


Fig. 1 Conservative vector fields for (a) sine curve $y = 2 \sin(1.4x) - 1$ and (b) star curve $\{(200 + 40 \sin(6\theta)) \cos \theta, (200 + 40 \sin(6\theta)) \sin \theta\}$. Note that the magnitudes of the vector fields are artificially set to be 20 for clarity (Similarly for the following vector fields)

coincides with the path following objective. Its gradient determines a conservative vector field for $\mathcal{C}_2$. Normalizing this gradient, we get a normalized conservative vector field as

$$\mathbf{v}_c = \frac{1}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (12)$$

Two conservative vector field examples are given in Fig. 1.

Tangential vector field of curve $\mathcal{C}_2$ can be chosen as the solenoidal vector field. Specifically, a normalized solenoidal vector field for curve $\mathcal{C}_2$ can be constructed as

$$\mathbf{v}_s = \frac{1}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} f_y \\ -f_x \end{pmatrix} \quad (13)$$

Solenoidal vector fields of the same two curves in Fig. 1 are illustrated in Fig. 2.

It can be found that the above two vector fields (12) and (13) are perpendicular to each other at every point in the 2D space, since we can easily verify that the inner product of $\mathbf{v}_c$ and $\mathbf{v}_s$ is zero. In fact, $\mathbf{v}_s$ aligns with the tangential direction of a point on the curve $\mathcal{C}_2$, while $\mathbf{v}_c$ is the normal direction of the same point.

2.3 Combined Vector Field

According to the Helmholtz decomposition, a certain combination of a conservative vector field and a solenoidal vector field can deduce a new vector field. Furthermore, recalling the characteristics of the two kinds of vector fields discussed above, if we can construct a vector field such that, for each point $\mathbf{p}$ in the 2D space,

- when $\mathbf{p}$ locates very far away from the desired curve $\mathcal{C}_2$ (i.e., $|r|$ is very large), its vector aligns with the conservative vector field (12) to direct straightly to $\mathcal{C}_2$,
- when $\mathbf{p}$ is on the curve $\mathcal{C}_2$ (i.e., $r = 0$), its vector aligns with the solenoidal vector field (13), i.e., along the desired path,
- the change of the vector field from $r \rightarrow \infty$ to $r = 0$ is continuous,

then the 2D curved path following can be guaranteed.

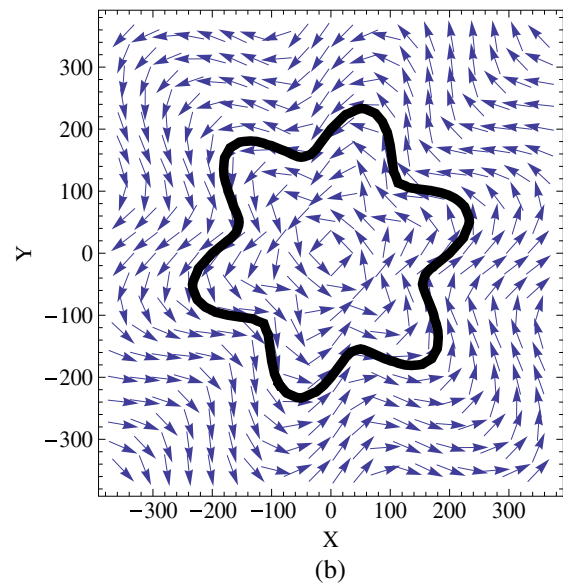
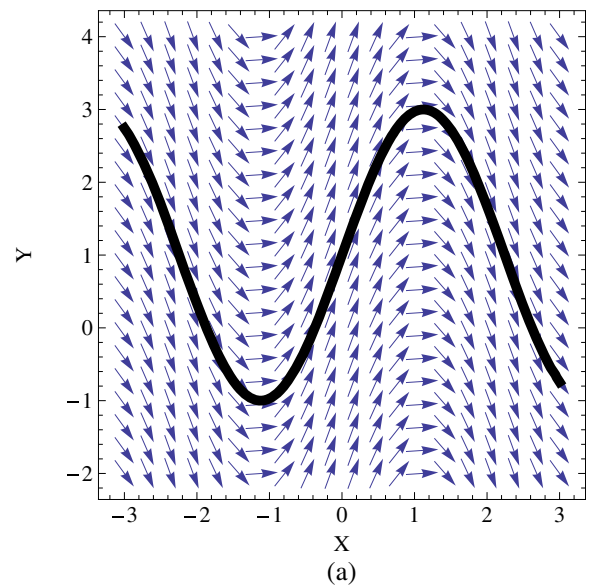


Fig. 2 Solenoidal vector fields for (a) sine curve $y = 2 \sin(1.4x) - 1$ and (b) star curve $\{(200 + 40 \sin(6\theta)) \cos \theta, (200 + 40 \sin(6\theta)) \sin \theta\}$

Following the previous concept, we combine (12) and (13) as the following form

$$\begin{aligned} \mathbf{v}_d = \begin{pmatrix} \dot{x}_d \\ \dot{y}_d \end{pmatrix} &= -V_g \tanh(\kappa r) \mathbf{v}_c + s V_g \operatorname{sech}(\kappa r) \mathbf{v}_s \\ &= \frac{V_g}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} -f_x \tanh(\kappa f(x, y)) + s f_y \operatorname{sech}(\kappa f(x, y)) \\ -f_y \tanh(\kappa f(x, y)) - s f_x \operatorname{sech}(\kappa f(x, y)) \end{pmatrix} \end{aligned} \quad (14)$$

where $\kappa > 0$ is a designing parameter. The larger κ is, the faster the convergence speed of the vector field to C_2 is, since in this case the conservative vector field plays a more important role. Parameter $s = \pm 1$ determines the direction of the vector field. For example, if the desired path is $C_2 : f(x, y) = x^2 + y^2 - 100 = 0$, which is a circle, then choosing $s = +1$ gives a clockwise vector field, while choosing $s = -1$ generates a counter-clockwise vector field. The magnitude of

Eq. 14 is set to be V_g to adapt the UAV's groundspeed. It can be easily verified that the three requirements listed above are satisfied. Examples of the constructed combined vector field for the two same curves in Fig. 1 are given in Fig. 3.

Theorem 1 *Reference trajectories determined by the combined vector field (14) will converge asymptotically to the desired curve C_2 .*

Proof Considering a Lyapunov function candidate $V_1 = r^2/2$ and evaluating its time derivative along this vector field, we have

$$\dot{V}_1 = r\dot{r} = -f \tanh(\kappa f) \sqrt{f_x^2 + f_y^2} \quad (15)$$

It can be easily seen that $\dot{V}_1 \leq 0$. Then the Lyapunov's direct method can be invoked to give us $r \rightarrow 0$, which implies the convergency of the reference trajectories determined by Eq. 14 to the desired curve C_2 . $\square$

Figure 4 shows us the reference trajectories determined by the combined vector field (14) for a star curve with three different κ choices. From the figure, we can observe the convergency of the reference trajectories, and additionally we see that larger κ implies faster convergence.

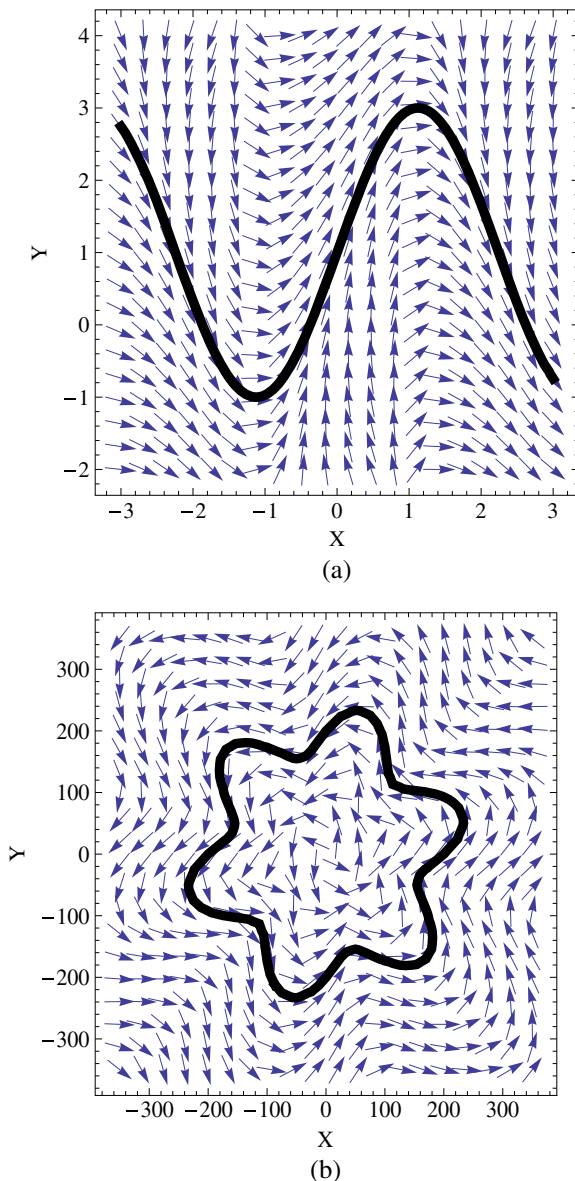


Fig. 3 Combined vector fields for (a) sine curve $y = 2 \sin(1.4x) - 1$ and (b) star curve $\{(200 + 40 \sin(6\theta)) \cos \theta, (200 + 40 \sin(6\theta)) \sin \theta\}$

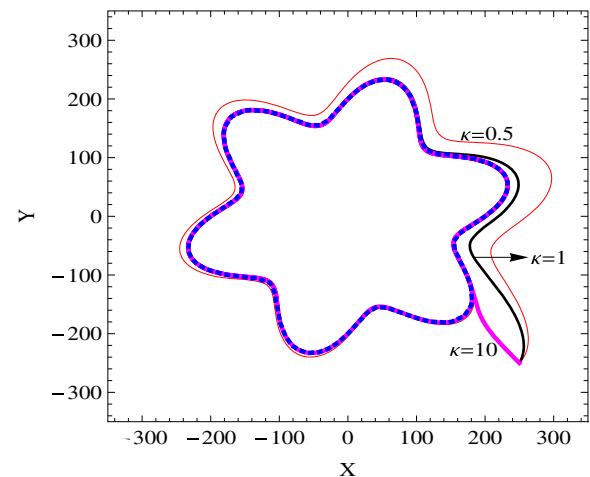


Fig. 4 Reference trajectories determined by the combined vector field (14) with three different parameter choices for the star curve $\{(200 + 40 \sin(6\theta)) \cos \theta, (200 + 40 \sin(6\theta)) \sin \theta\}$ initiating from (250, -250)

2.4 Controller Design

If no input constraint exists for the UAV flight, then the reference trajectories determined by the combined vector field (14) can be exactly tracked to achieve the curved path following. However in reality it is not the case, and a maximum course rate constraint always exists, i.e., the above-mentioned $\dot{\chi}_{\max}$. In the following we will design a saturated course rate controller for constrained UAVs based on the above-constructed combined vector field.

The desired course determined by the combined vector field (14) can be computed as

$$\chi_d = \arctan2(\dot{y}_d, \dot{x}_d) = \arctan2(-f_y \tanh(\kappa f) - s f_x \operatorname{sech}(\kappa f), -f_x \tanh(\kappa f) + s f_y \operatorname{sech}(\kappa f)) \quad (16)$$

where $\arctan2(\cdot, \cdot)$ is the four-quadrant inverse tangent function. Denote the course error between the UAV and the combined vector field (14) as $\chi_e = \langle \chi - \chi_d \rangle \in (-\pi, \pi]$. Here $\langle \cdot \rangle$ is used to wrap the angles. Taking this error into account, the time derivative of χ_d can be computed as

$$\dot{\chi}_d = \frac{\partial \chi_d}{\partial x} \dot{x} + \frac{\partial \chi_d}{\partial y} \dot{y} = V_g (\mathcal{A}_x \cos \chi + \mathcal{A}_y \sin \chi) \quad (17)$$

where

$$\mathcal{A}_x = \frac{\partial \chi_d}{\partial x} = -s \kappa f_x \operatorname{sech}(\kappa f) - (f_{xx} f_y - f_x f_{xy}) / (f_x^2 + f_y^2) \quad (18)$$

$$\mathcal{A}_y = \frac{\partial \chi_d}{\partial y} = -s \kappa f_y \operatorname{sech}(\kappa f) + (f_{yy} f_x - f_y f_{xy}) / (f_x^2 + f_y^2) \quad (19)$$

We can now design a saturated course rate controller for the path following of curve $\mathcal{C}_2$ as follows

$$u_1 = \begin{cases} \omega_1, & \text{if } |\omega_1| \leq \dot{\chi}_{\max} \\ \operatorname{sgn}(\omega_1) \dot{\chi}_{\max}, & \text{if } |\omega_1| > \dot{\chi}_{\max} \end{cases} \quad (20)$$

where $\operatorname{sgn}(\cdot)$ is the sign function, and

$$\omega_1 = -k_\chi \chi_e + \dot{\chi}_d \quad (21)$$

with k_χ being a positive feedback gain.

Remark 1 If we do not introduce the course error χ_e into $\dot{\chi}_d$, i.e., if instead of (17), we use

$$\dot{\chi}_d = \frac{\partial \chi_d}{\partial x} \dot{x}_d + \frac{\partial \chi_d}{\partial y} \dot{y}_d \quad (22)$$

then the path following convergence will slow down, or in the worst case, the path following convergence cannot be ensured. This was first found and discussed in detail in [18] for circular path following.

Based on the desired course rate calculated by Eq. 17, we give the following two definitions for the combined vector field (14) regarding the 2D desired curve $\mathcal{C}_2$.

Definition 1 We say that the combined vector field (14) is globally feasible for $\mathcal{C}_2$ if $|\dot{\chi}_d| \leq \dot{\chi}_{\max}$ holds at each point of the 2D space.

Definition 2 We say that the combined vector field (14) is locally feasible for $\mathcal{C}_2$ if $|\dot{\chi}_d| \leq \dot{\chi}_{\max}$ holds just in a domain that encloses $\mathcal{C}_2$.

Remark 2 Because it can be easily seen from Eq. 17 that the desired course rate $\dot{\chi}_d$ changes continuously, similarly as the initial loiter circle discussed in [18], the UAV course will align with that of the combined vector field after an initial alignment process, no matter what the initial course error χ_e is. The global feasibility of the vector field guarantees that no matter what the initial absolute distance $|r_0|$ is, the UAV can exactly fly along the reference trajectory determined by the combined vector field (14) if its initial course has been aligned.

From Eq. 17, we can see that whether a combined vector field with form Eq. 14 is feasible (globally or locally) or not depends not only on the vector field parameters but also on the desired curve $\mathcal{C}_2$. Given a course rate constraint $\dot{\chi}_{\max}$, there exists a feasible condition for Eq. 14 to be a globally feasible vector field. The condition is given in the following theorem with the proof given in Appendix A.

Theorem 2 A feasible condition for Eq. 14 to be a globally feasible vector field concerning the desired curve $\mathcal{C}_2$ is that, for each point in the 2D space, we have

$$\sqrt{\mathcal{A}_x^2 + \mathcal{A}_y^2} \leq \frac{7\dot{\chi}_{\max}}{10V_g} \quad (23)$$

Remark 3 We can use condition (23) to choose an appropriate parameter κ for the combined vector field (14). As an illustrative example, consider a straight

line given by $f(x, y) = ax + by + c = 0$. In this situation, $\mathcal{A}_x = -s\kappa a \operatorname{sech}(\kappa f)$ and $\mathcal{A}_y = -s\kappa b \operatorname{sech}(\kappa f)$. Substituting them into Eq. 23, we have

$$\sqrt{(-s\kappa a \operatorname{sech}(\kappa f))^2 + (-s\kappa b \operatorname{sech}(\kappa f))^2} \leq \frac{7\dot{\chi}_{\max}}{10V_g} \quad (24)$$

Then κ can be chosen to satisfy $\kappa \leq 7\dot{\chi}_{\max} / (10V_g \sqrt{a^2 + b^2})$.

Remark 4 Condition (23) also puts forward a scale requirement for the curve function $f(x, y)$. For example, consider the same straight line given in Remark 3 but increase the scale of $f(x, y)$ as $f(x, y) = 10ax + 10by + 10c$. Then the κ chosen in Remark 3 may no longer satisfy the feasible condition, since in this case, the feasible condition becomes

$$10\sqrt{(-s\kappa a \operatorname{sech}(\kappa f))^2 + (-s\kappa b \operatorname{sech}(\kappa f))^2} \leq \frac{7\dot{\chi}_{\max}}{10V_g} \quad (25)$$

Remark 5 We can see from the proof that in fact, we just need $|\cos \chi_{e0}|$ in Eq. A.5 is not more than $(\sqrt{2}/2 - \epsilon)$, where ϵ is a nonzero small positive number. For simplicity we choose an approximate fraction 7/10.

Remark 6 In reality it is impossible for a UAV to fly all over the whole 2D space. The UAV's flight area is actually bounded. Therefore we only need the satisfaction of condition (23) in a bounded domain that encloses the desired curve and the UAV's maximum possible flight area. That is, we only need that the combined vector field is locally feasible.

2.5 Stability Analysis

Firstly we can derive the following result from Theorem 2, the proof of which is given in Appendix B.

Lemma 1 Given controller (20) and condition (23), there exists a positive constant k_2 such that

$$\begin{cases} \dot{\chi}_e \leq -k_2 \sin \chi_e, & \text{if } \chi_e \in (0, \pi] \\ \dot{\chi}_e \geq -k_2 \sin \chi_e, & \text{if } \chi_e \in (-\pi, 0] \end{cases} \quad (26)$$

Remark 7 We can see from (26) that $\chi_e \dot{\chi}_e \leq -k_2 \chi_e \sin \chi_e \leq 0$. This implies that regardless of the original value, the absolute value of χ_e decreases with time under controller (20). This further means that the UAV course will gradually reach the desired course determined by the combined vector field.

Another two lemmas are given as below before we discuss the Lyapunov stability of the saturated course rate controller (20). The proofs of these two lemmas are given in Appendices C and D respectively.

Lemma 2 Assume that $\sqrt{f_x^2 + f_y^2} \leq M$ with $M > 0$. Given controller (20) and condition (23), we have that $r = f(x, y)$ is bounded, i.e., there exists a positive constant K such that $|r| \leq K$.

Lemma 3 If $0 < \rho \leq \kappa \operatorname{sech}^2(\kappa K)/2$, then we have $x \tanh(\kappa x) \geq \rho x^2$ for all $0 \leq x \leq K$.

The main result about the Lyapunov stability of controller (20) is summarized in the following theorem.

Theorem 3 Considering the UAV kinematics system (5)–(7), the saturated course rate controller given in Eq. 20 and the condition given in Eq. 23, the UAV trajectory asymptotically converges to the desired curve $\mathcal{C}_2$. That is to say, as the time $t \rightarrow \infty$, the directional distance r converges to zero, and the course error between the UAV trajectory and the desired curve converges to zero as well.

Proof It needs to be pointed out that the proof is based on the combined vector field parameter choice $s = +1$, and the proof when $s = -1$ is similar.

Let us consider the Lyapunov function candidate

$$V_2 = \frac{1}{2}r^2 + \frac{\lambda}{2}\chi_e^2 \quad (27)$$

where $\lambda > 0$ is an adjustable scaling factor. Its time derivative is given by

$$\begin{aligned} \dot{V}_2 &= r\dot{r} + \lambda\chi_e\dot{\chi}_e \leq fV_g\sqrt{f_x^2 + f_y^2} \sin(\chi_e + \chi_r) \\ &\quad - \lambda k_2 \chi_e \sin \chi_e \end{aligned} \quad (28)$$

When $\chi_e = \pi$, it can be seen from Eqs. A.11 and A.12 that $\dot{\chi}_e \neq 0$, which implies that χ_e will fall into $(-\pi, \pi)$. Thus we just consider the situation when

$\chi_e \in (-\pi, \pi)$. The discussion is partitioned into two cases.

Case 1. $|\chi_e| \in (\pi/2, \pi)$

In this case it can be easily found that $\chi_e \sin \chi_e \geq \pi/2 |\sin \chi_{e0}|$. Thereby if we take $\lambda \geq 2KM(V_a + W)/(\pi k_2 |\sin \chi_{e0}|)$, then we have

$$\dot{V}_2 \leq KM(V_a + W) - \frac{\pi \lambda k_2}{2} |\sin \chi_{e0}| \leq 0 \quad (29)$$

Case 2. $\chi_e \in [-\pi/2, \pi/2]$

In this case we have $\chi_e \sin \chi_e \geq \sin^2 \chi_e \geq \sin^2(\chi_e/2) \geq 0$. This yields

$$\begin{aligned} \dot{V}_2 &\leq fV_g \sqrt{f_x^2 + f_y^2} \left[2 \sin\left(\frac{\chi_e}{2}\right) \cos\left(\frac{\chi_e}{2}\right) \cos \chi_r + \left(1 - 2 \sin^2\left(\frac{\chi_e}{2}\right)\right) \sin \chi_r \right] - \lambda k_2 \sin^2\left(\frac{\chi_e}{2}\right) \\ &= fV_g \sqrt{f_x^2 + f_y^2} \sin \chi_r + 2fV_g \sqrt{f_x^2 + f_y^2} \sin\left(\frac{\chi_e}{2}\right) \cos\left(\chi_r + \frac{\chi_e}{2}\right) - \lambda k_2 \sin^2\left(\frac{\chi_e}{2}\right) \\ &\leq -V_g |f| \sqrt{f_x^2 + f_y^2} \tanh(\kappa |f|) + 2V_g |f| \sqrt{f_x^2 + f_y^2} \left| \sin\left(\frac{\chi_e}{2}\right) \right| - \lambda k_2 \left| \sin\left(\frac{\chi_e}{2}\right) \right|^2 \\ &\leq -\sqrt{f_x^2 + f_y^2} \left[V_g \rho |f|^2 - 2V_g |f| \left| \sin\left(\frac{\chi_e}{2}\right) \right| + \frac{\lambda k_2}{\sqrt{f_x^2 + f_y^2}} \left| \sin\left(\frac{\chi_e}{2}\right) \right|^2 \right] \\ &\leq -\sqrt{f_x^2 + f_y^2} \left[(V_a - W) \rho |f|^2 - 2(V_a + W) |f| \left| \sin\left(\frac{\chi_e}{2}\right) \right| + \frac{\lambda k_2}{\sqrt{f_x^2 + f_y^2}} \left| \sin\left(\frac{\chi_e}{2}\right) \right|^2 \right] \end{aligned} \quad (30)$$

Here the fact given in Lemma 3 is used. It can be seen from Eq. 30 that, if we take $\lambda \geq M(V_a + W)^2/(\rho k_2(V_a - W))$, then we have

$$\begin{aligned} \dot{V}_2 &\leq -\sqrt{f_x^2 + f_y^2} \left[\sqrt{(V_a - W) \rho} |f| \right. \\ &\quad \left. - \frac{\sqrt{\lambda k_2}}{(f_x^2 + f_y^2)^{1/4}} \left| \sin\left(\frac{\chi_e}{2}\right) \right| \right]^2 \leq 0 \end{aligned} \quad (31)$$

We know from Eq. 15 that r decays to 0 under the combined vector field (14). Recalling that “ $s = +1$ ” is considered, $r \rightarrow 0$ further indicates that

$$\mathbf{v}_d \rightarrow \frac{V_g}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} f_y \\ -f_x \end{pmatrix} \quad (32)$$

This yields $\chi_d \rightarrow \chi_p$. Here χ_p is the tangential direction of the curve corresponding to the current UAV location (See Appendix B for detail).

For the Lyapunov function candidate V_2 given by Eq. 27, since $\dot{V}_2 \leq 0$ is guaranteed by Eqs. 29 and 31,

we conclude that $r \rightarrow 0$ and $\chi_e \rightarrow 0$ as $t \rightarrow +\infty$. $\chi_e \rightarrow 0$ means $\chi \rightarrow \chi_d$, which further leads to $\chi \rightarrow \chi_p$ by combining the conclusion $\chi_d \rightarrow \chi_p$. This ends the proof of this theorem. $\square$

Remark 8 The proof of Theorem 3 is based on the condition that the combined vector field (14) is globally feasible. In fact we just need that Eq. 14 is locally feasible in a domain $\mathcal{D}_2$, which satisfies the following two conditions: (i) it encloses $\mathcal{C}_2$, (ii) the cross-track distances of the points on its lateral bound is not less than the bound K given in Lemma 2. For example, if the straight line in Remark 3 is again considered, we just need that Eq. 14 is locally feasible in the domain between the two lines $ax + by + c = K$ and $ax + by + c = -K$.

3 3D Curved Path Following

3.1 Problem Formulation

In this section we investigate the 3D curved path following problem. Similarly as for the 2D case, we introduce the course χ and the climb angle γ ,

and we use the following simplified 3D kinematics model for the UAV which is modeled in inertial XYZ frame

$$\dot{x} = V_g \cos \gamma \cos \chi \quad (33)$$

$$\dot{y} = V_g \cos \gamma \sin \chi \quad (34)$$

$$\dot{z} = V_g \sin \gamma \quad (35)$$

$$\dot{\chi} = u_1 \quad (36)$$

$$\dot{\gamma} = u_2 \quad (37)$$

$$V_g = (W_x \cos \chi + W_y \sin \chi) \cos \gamma + \frac{1}{2} \sqrt{4V_a^2 - 3(W_x^2 + W_y^2) + (1 + 2 \cos(2\gamma))(W_x \cos \chi + W_y \sin \chi)^2 - (W_x \sin \chi - W_y \cos \chi)^2} \quad (39)$$

Given a desired 3D arbitrary twice differentiable curved path $\mathcal{C}_3$ which is determined by $\{f(x, y, z) = 0, g(x, y, z) = 0\}$, we define a distance $r = \sqrt{f^2(\mathbf{p}) + g^2(\mathbf{p})} = \sqrt{f^2(x, y, z) + g^2(x, y, z)}$ to measure the closeness of the current UAV position $\mathbf{p}$ to the desired curve $\mathcal{C}_3$.

Then the 3D arbitrary twice differentiable curved path following problem can be summarized as: To design input commands u_1 and u_2 for a fixed-wing UAV whose kinematics is given by Eqs. 33–37 such that (i) the input constraints (38) holds, and (ii) if possible, make the UAV fly along the 3D desired curve $\mathcal{C}_3$, and if not, make the distance r as small as we can.

3.2 Combined Vector Field

Similarly as for the 2D curved path following case, we first construct a conservative vector field and a solenoidal vector field for $\mathcal{C}_3$. Then based on the Helmholtz theorem, we construct a combined vector field by trading off these two vector fields.

Considering a potential function $f_c = r^2 = f^2(x, y, z) + g^2(x, y, z)$ and normalizing its gradient, we get a conservative vector field for $\mathcal{C}_3$ as follows

$$\mathbf{v}_c = \frac{1}{N_c} \begin{pmatrix} ff_x + gg_x \\ ff_y + gg_y \\ ff_z + gg_z \end{pmatrix} \quad (40)$$

where $\mathbf{p} = (x, y, z)^T$ is the UAV position, and $\mathbf{v} = (\dot{x}, \dot{y}, \dot{z})^T$ denotes the inertial velocity. Same as before, $V_g = \|\mathbf{v}\|$ is the groundspeed, $\chi \in (-\pi, \pi]$ is the UAV course, and u_1 is the course rate input. γ denotes the climb angle with u_2 being the commanded climb rate. Input constraints are given as

$$|u_1| \leq \dot{\chi}_{\max}, |u_2| \leq \dot{\gamma}_{\max} \quad (38)$$

The relationship between the airspeed and the ground-speed in 3D space is

where $N_c = \sqrt{(ff_x + gg_x)^2 + (ff_y + gg_y)^2 + (ff_z + gg_z)^2}$ is the normalization factor. Similarly the following tangential vector field for curve $\mathcal{C}_3$

$$\mathbf{v}_s = \frac{1}{N_s} \begin{pmatrix} f_y g_z - f_z g_y \\ f_z g_x - f_x g_z \\ f_x g_y - f_y g_x \end{pmatrix} \quad (41)$$

where $N_s = \sqrt{(f_y g_z - f_z g_y)^2 + (f_z g_x - f_x g_z)^2 + (f_x g_y - f_y g_x)^2}$ is the normalization factor, can be chosen as a solenoidal vector field.

According to the combination concept given for 2D combined vector field, we construct a 3D combined vector field as

$$\begin{aligned} \mathbf{v}_d = \begin{pmatrix} \dot{x}_d \\ \dot{y}_d \\ \dot{z}_d \end{pmatrix} &= -V_g \tanh(\kappa r) \mathbf{v}_c + s V_g \operatorname{sech}(\kappa r) \mathbf{v}_s \\ &= \begin{pmatrix} -\frac{V_g(ff_x + gg_x) \tanh(\kappa r)}{N_c} + \frac{s V_g(f_y g_z - f_z g_y) \operatorname{sech}(\kappa r)}{N_s} \\ -\frac{V_g(ff_y + gg_y) \tanh(\kappa r)}{N_c} + \frac{s V_g(f_z g_x - f_x g_z) \operatorname{sech}(\kappa r)}{N_s} \\ -\frac{V_g(ff_z + gg_z) \tanh(\kappa r)}{N_c} + \frac{s V_g(f_x g_y - f_y g_x) \operatorname{sech}(\kappa r)}{N_s} \end{pmatrix} \end{aligned} \quad (42)$$

Similarly as those in the 2D combined vector field (14), here $s = \pm 1$ determines the path following direction, and $\kappa > 0$ is a designing parameter. It can be found that if $\mathbf{p}$ is very far away, which implies that $\operatorname{sech}(\kappa r)$ is nearly zero, the direction determined by Eq. 42 points almost directly towards $\mathcal{C}_3$. And if $\mathbf{p}$ is

on the desired path, the direction determined by Eq. 42 aligns with the tangential direction of C_3 .

Theorem 4 *Reference trajectories determined by the 3D combined vector field (42) will converge asymptotically to the 3D desired curve C_3 .*

Proof Consider the same Lyapunov function candidate $V_1 = r^2/2$. Simplifying its time derivative along this combined vector field yields

$$\dot{V}_1 = -N_c V_g \tanh(\kappa r) \quad (43)$$

It can be easily seen that $\dot{V}_1 \leq 0$, since for the 3D curved path following, the defined distance r is a nonnegative value. Then from the Lyapunov's direct method, we can see that $r \rightarrow 0$, which implies the convergency of the reference trajectories determined by Eq. 42 to the desired curve C_3 . $\square$

Considering a 3D star curve whose parametric equation is given by

$$\begin{cases} x = (200 + 20 \sin(8\theta)) \left(\frac{\cos \theta}{\sqrt{2}} + \frac{3 \sin \theta}{\sqrt{22}} \right) \\ y = (200 + 20 \sin(8\theta)) \left(-\frac{\cos \theta}{\sqrt{2}} + \frac{3 \sin \theta}{\sqrt{22}} \right) \\ z = -\frac{\sqrt{2}}{\sqrt{11}} (200 + 20 \sin(8\theta)) \sin \theta \end{cases} \quad (44)$$

the reference trajectories determined by the combined vector field (42) are shown in Fig. 5, which gives us an illustrative verification of Theorem 4.

3.3 Controller Design

In this subsection, based on the constructed combined vector field (42), we design a jointly saturated course rate and climb rate controller for the 3D curved path following by taking the input constraints (38) into consideration.

Firstly we compute the desired course and the desired climb angle determined by Eq. 42 as follows

$$\chi_d = \arctan2(\dot{y}_d, \dot{x}_d) \quad (45)$$

$$\gamma_d = \arctan2\left(\dot{z}_d, \sqrt{\dot{x}_d^2 + \dot{y}_d^2}\right) \quad (46)$$

Denote the course error as $\chi_e = \chi - \chi_d \in (-\pi, \pi]$ and the climb angle error as $\gamma_e = \gamma - \gamma_d \in (-\pi, \pi]$. Similarly by taking these two flight path

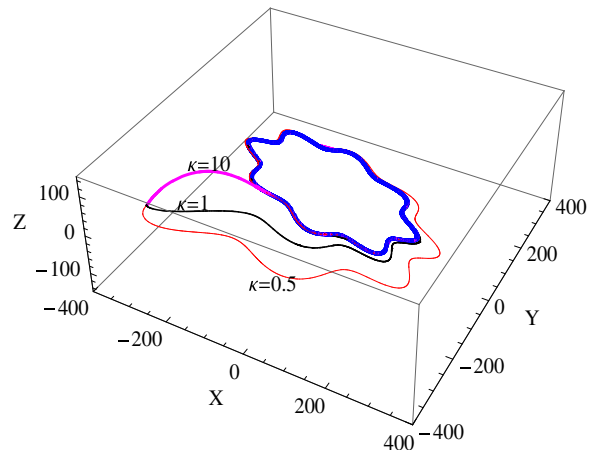


Fig. 5 Reference trajectories determined by the combined vector field (42) with three different parameter choices for the 3D star curve with parametric Eq. 44 initiating from (250,-250,0)

angle errors into account, we can then calculate the time derivatives of the desired flight path angles as

$$\dot{\chi}_d = \frac{\partial \chi_d}{\partial x} \dot{x} + \frac{\partial \chi_d}{\partial y} \dot{y} + \frac{\partial \chi_d}{\partial z} \dot{z} = V_g (B_x \cos \gamma \cos \chi + B_y \cos \gamma \sin \chi + B_z \sin \gamma) \quad (47)$$

$$\dot{\gamma}_d = \frac{\partial \gamma_d}{\partial x} \dot{x} + \frac{\partial \gamma_d}{\partial y} \dot{y} + \frac{\partial \gamma_d}{\partial z} \dot{z} = V_g (C_x \cos \gamma \cos \chi + C_y \cos \gamma \sin \chi + C_z \sin \gamma) \quad (48)$$

where B_x , B_y , B_z , C_x , C_y and C_z can be computed by the chain rule and omitted here for simplicity. The jointly saturated course rate and climb rate controller is designed as

$$u_1 = \begin{cases} \omega_1, & \text{if } |\omega_1| \leq \dot{\chi}_{\max} \\ \text{sgn}(\omega_1) \dot{\chi}_{\max}, & \text{if } |\omega_1| > \dot{\chi}_{\max} \end{cases} \quad (49)$$

$$u_2 = \begin{cases} \omega_2, & \text{if } |\omega_2| \leq \dot{\gamma}_{\max} \\ \text{sgn}(\omega_2) \dot{\gamma}_{\max}, & \text{if } |\omega_2| > \dot{\gamma}_{\max} \end{cases} \quad (50)$$

where $\omega_1 = -k_\chi \chi_e + \dot{\chi}_d$ and $\omega_2 = -k_\gamma \gamma_e + \dot{\gamma}_d$. k_χ and k_γ are the feedback gains.

Similarly we give the following two definitions for the 3D combined vector field (42) relating to the 3D desired curve C_3 .

Definition 3 We say that the 3D combined vector field (42) is globally feasible for C_3 if both $|\dot{\chi}_d| \leq \dot{\chi}_{\max}$ and $|\dot{\gamma}_d| \leq \dot{\gamma}_{\max}$ hold at each point of the 3D space.

Definition 4 We say that the 3D combined vector field (42) is locally feasible for C_3 if both $|\dot{\chi}_d| \leq \dot{\chi}_{\max}$ and $|\dot{\gamma}_d| \leq \dot{\gamma}_{\max}$ hold just in a domain that encloses C_3 .

Remark 9 Similarly as for the 2D case, the global feasibility of the vector field guarantees that no matter what the initial distance r_0 is, the UAV can exactly track the reference trajectory determined by the 3D combined vector field (42) if its initial flight path angles have been aligned.

The feasibility of the 3D combined vector field (42) also depends on both the vector field parameters and the desired curve C_3 , with the feasibility condition summarized in the following theorem.

Theorem 5 A feasible condition for Eq. 42 to be a globally feasible vector field concerning the desired curve C_3 is that, for each point in the 3D space, we have

$$\sqrt{C_x^2 + C_y^2 + C_z^2} \leq \frac{7\dot{\gamma}_{\max}}{10V_g} \quad (51a)$$

$$\sqrt{B_x^2 + B_y^2} \leq \frac{7(1-\alpha)\dot{\chi}_{\max}}{10V_g}, \quad |B_z| \leq \frac{\alpha\dot{\chi}_{\max}}{V_g} \quad (51b)$$

with $0 < \alpha < 1$.

The proof of this theorem is given in Appendix E.

Remark 10 Similarly as discussed in Remark 5, condition (51) is a conservative condition to ensure the global feasibility of the combined vector field (42). The condition relies on both of the combined vector field parameters and the desired curve C_3 . And similarly as explained in Remarks 6 and 8, local feasibility of Eq. 42 in a domain that encloses the 3D desired curve C_3 and the UAV's maximum possible flight area is enough to achieve the 3D curved path following.

4 Approach Assessment

We have now proposed two approaches respectively for the 2D curved path following and the 3D curved

path following in the framework of guidance vector field approach.

The advantages of the proposed approaches are four-fold.

- Theoretically the approaches can be used to follow arbitrary 2D and 3D twice differentiable curves, including straight lines and planar circles considered mostly in the literature.
- A common parameter κ , which can be used to tune the convergence speed of the path following, and another binary common parameter s , which can be used to change the path following direction, are included. These two parameters provide the proposed approaches with some adjustability.
- Compared with the 2D curved path following approach given in [17], the proposed approach explicitly addresses the course rate saturation, suffers no chattering problem, includes actually only one designing parameter since the path following direction has always been chosen before the path following mission, and need not seek reference points on the desired curve.
- Same as the vector field approach surveyed in [13], the proposed approaches make the UAV follow the desired path accurately, require little control effort and are easy to be implemented.

In addition several limitations of the approaches should be pointed out.

- No common parameter κ exists for all the twice differentiable curves. That is, for different curves, we should choose different suitable κ 's, which are greatly different from each other. This is because the two coefficients used to trade off the conservative vector field and the solenoidal vector field, $\tanh(\kappa r)$ and $\text{sech}(\kappa r)$, are not scale-free. Their scales differ a lot for different curves. One strategy to alleviate this is introducing an alternative parameter κ_0 and setting $\kappa = \kappa_0/|r_0|$ where $r_0 \neq 0$ is the initial distance.
- For a certain 3D desired curve, because unlike its unique parametric representation, there are infinite number of function pairs (i.e., $\{f(x, y, z), g(x, y, z)\}$'s) to represent it, problems may well emerge if the scales of $f(x, y, z)$ and $g(x, y, z)$ differ greatly. For example, in the considered domain, if the scale of $f(x, y, z)$ is

10^{15} and the scale of $g(x, y, z)$ is 1, then the approach may steer the UAV to the surface determined by $f(x, y, z) = 0$ instead of the 3D curve determined by $\{f(x, y, z) = 0, g(x, y, z) = 0\}$. This is because the distance is defined as $r = \sqrt{f^2 + g^2}$, and due to this, the approach may mistakenly think that the UAV has already been on the surface determined by $g(x, y, z) = 0$. That is to say, the 3D curved path following result relies significantly on the scales of the two chosen curve functions. One strategy to alleviate this is to normalize the curve functions in the considered domain, i.e., to make their maximum value in this domain be 1. An indirect strategy is to redefine the distance r as $r = \sqrt{f^2 + (\lambda g)^2}$ by introducing a scaling factor λ and then redesign the combined vector field approach. For a certain 2D desired curve, given an arbitrary nonzero scaling factor λ , replacing $f(x, y)$ by $\lambda f(x, y)$ does not affect the path following result when $\kappa = \kappa_0/|r_0|$ is chosen. This is because this scaling factor can be eliminated for both χ_d and $\dot{\chi}_d$, which can be easily seen from Eqs. 16–19. However for a same κ_0 , two different choices for $f(x, y)$ not with a scaling factor relationship may result in different convergence speed.

- The proposed path following approaches do not seek reference points on the desired curve, and thus they are not easy to be extended to the trajectory tracking problem compared with the reference point based approaches, such as the approach given in [17].

5 Simulation Results

In this section several simulation studies in both 2D and 3D spaces are designed to see the performances of the proposed path following approaches. Note that in the recent survey paper [13], the authors have systematically compared the performances of five main different path following approaches (the guidance vector field approach included) for 2D straight lines and 2D circles. Focusing on the capability of following arbitrary twice differentiable curves, we just give the performance comparison of the proposed 2D curved path following approach with that given in [17]. And since no 3D curved path following approach has been reported in the literature, 3D simulations are just given

to see the effectiveness and the limitations of the proposed 3D curved path following approach.

We set the sampling frequency of the controller as 10 Hz. The constant airspeed of the UAV is set to be $V_a = 20$ m/s, and a constant wind disturbance with velocity $\mathbf{W} = [6, 8, 0]$ m/s is assumed to be present, which means that the wind speed accounts for 50 percent of the UAV airspeed. Course rate and climb rate constraints are set to be $\dot{\chi}_{\max} = 0.5$ rad/s and $\dot{\gamma}_{\max} = 0.3$ rad/s respectively. The feedback gains, k_χ and k_γ , are both chosen to be 1. Assuming that the initial distance r_0 ($r_0 = f_0$ for 2D case, and $r_0 = \sqrt{f_0^2 + g_0^2}$ for 3D case) is not zero, we define the relative distance as r/r_0 . The strategy $\kappa = \kappa_0/|r_0|$ is used in the simulations.

5.1 2D Case: Parameter Analysis

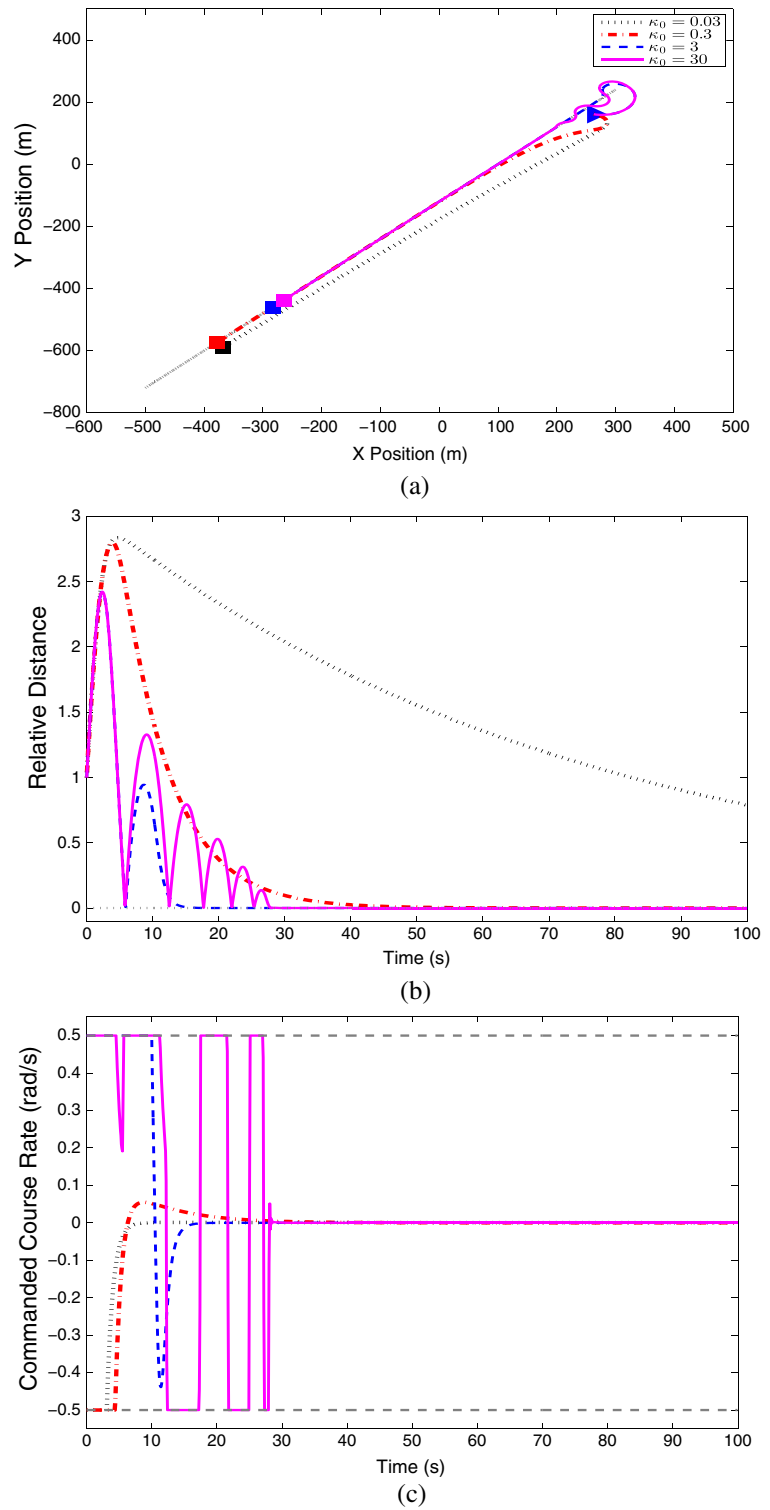
Consider a 2D straight line path determined by $f(x, y) = -360x + 300y + 36000 = 0$. The UAV is assumed to be initially located at (263, 161) with course 5.97. The alternative parameter κ_0 is set to be four different constants, 0.03, 0.3, 3 and 30. The corresponding path following results are shown in Fig. 6.

We can conclude that if κ_0 is too small, then the path following convergence is very slow (See the result when $\kappa_0 = 0.03$). The reason is that in this case the weighted conservative component of the combined vector field is not large enough for rapid approaching to the desired path. As κ_0 becomes larger, the path following convergence becomes faster (See the results when $\kappa_0 = 0.3$ and 3). However if κ_0 is too large, the desired course rate (16) may exceed the UAV course rate constraint during a long period, and this further results in some repeated UAV trajectory oscillations around the desired path before the convergence (See the result when $\kappa_0 = 30$), or even worse, infinite repeated UAV trajectory oscillations around the desired path with no convergence (e.g., when $\kappa_0 = 300$).

5.2 2D Case: Performance Comparison

In this subsection, we compare the proposed approach with that given in Ref. [17] through three simulation examples. The initial courses of the UAVs are all set to be 5.97. For the proposed approach, we set $\kappa_0 = 1$. For the approach proposed in Ref. [17], to

Fig. 6 Path following results of the 2D straight line $f(x, y) = -360x + 300y + 36000 = 0$: **a** UAV trajectory; **b** Relative distance; **c** Commanded course rate. Initial UAV position is denoted as a filled triangle, and terminal UAV position is denoted as a filled square (Similarly hereinafter)



avoid the probable over-frequent input saturations, we choose a comparatively large width for the sliding mode boundary region, specifically we set $\epsilon = 0.1$, and additionally we set

$$\chi^\infty = \pi/4, \quad k = 0.1, \quad \rho = 100, \quad \bar{d} = 10 \quad (52)$$

$$\mu = \max \left\{ \frac{2(1 + (k\bar{d})^2)}{\cos\left(\frac{2\chi^\infty}{\pi} \arctan(k\bar{d})\right)}, \frac{4\chi^\infty k\bar{d}}{\pi \sin\left(\frac{2\chi^\infty}{\pi} \arctan(k\bar{d})\right)} \right\} \quad (53)$$

$$\beta = \frac{V_g \bar{d}}{2\rho} \quad (54)$$

Although under these parameter settings, the condition given by (13) in Ref. [17] does not hold, we find from the following discussions that the undesired input saturations can be effectively avoided.

In the first example, we consider an ellipse with equation $f(x, y) = x^2/250^2 + y^2/200^2 - 1 = 0$. The initial position of the first two UAVs is (150, -50), and that of the last two UAVs is (250, -200). The path following results are given in Fig. 7. In the second example, two sine curves determined by $f(x, y) = 500 \sin((x - 800)/p) + 300 - y = 0$ with $p = 100$ and $p = 170$ respectively are chosen as the desired curved paths. The first two UAVs initially locate at (757, 298) and are set to follow the sine curve with $p = 100$, while the last two UAVs initially locate at (-757, 298) and are set to follow the sine curve with $p = 170$. The path following comparisons are illustrated in Fig. 8. In the third example, a 2D star curve determined by

$$f(x, y) = \frac{1}{(x^2 + y^2)^{9/2} + 1} [-a_0 x^8 - 8b_0 x^7 y - 4a_0 x^6 y^2 + 56b_0 x^5 y^3 - 6a_0 x^4 y^4 - 56b_0 x^3 y^5 - 4a_0 x^2 y^6 + 8b_0 x y^7 - a_0 y^8 + (x^8 + x^6 y^2 + x^4 y^4 + x^2 y^6 + y^8) \sqrt{x^2 + y^2}] = 0 \quad (55)$$

with $a_0 = 200$ and $b_0 = 10$ is considered. The term $1/((x^2 + y^2)^{9/2} + 1)$ is artificially added to change the scale of the curve function. If this term is not added, comparable path following convergence speed can be obtained by choosing $\kappa_0 = 10$. The result is not given for simplicity. The initial positions of the first two and the last two UAVs are set to be (250, -250) and (-250, -250) respectively. The results are shown in Fig. 9.

It can be seen that all of the desired example curves can be followed by the two approaches, and after the

initial adjustment periods, the path following results are very similar for the two approaches. We can then conclude that their path following performances are comparable. However it should be pointed out that only one parameter needs to be designed for our proposed approach, and by contrast, up to 7 parameters should be set for the approach given in Ref. [17].

5.3 3D Case: Limitation Analysis

In this subsection, we consider a 3D straight line determined by

$$\begin{aligned} f(x, y, z) &= -300x - 800y = 0, \\ g(x, y, z) &= 200y + 300z = 0 \end{aligned} \quad (56)$$

In the considered domain $[-300, 2000] \times [-700, 500] \times [-100, 300]$, the maximum value of the two functions are about 10^6 and 10^5 respectively, which implies comparable scales for them. To see the influence of the scales of the curve functions on the path following performance, we multiply $g(x, y, z)$ by an scale factor 10^5 , i.e., we consider the same straight line determined by

$$\begin{aligned} f(x, y, z) &= -300x - 800y = 0, \\ g(x, y, z) &= 10^5(200y + 300z) = 0 \end{aligned} \quad (57)$$

The UAVs are assumed to be initially located at (-120, 20, 20) with initial course -0.889 and initial climb angle 1.028. Choosing four values for κ_0 , 0.03, 0.3, 3 and 30, the path following results of this 3D straight line determined respectively by Eqs. 56 and 57 are shown in Figs. 10 and 11 respectively.

We can see from Fig. 10 that when the two curve functions have comparable scales, except that the convergence speed is very slow for the parameter setting $\kappa_0 = 0.03$, all of the parameter settings $\kappa_0 = 0.3, 3$ and 30 result in path following convergence to the desired 3D straight line. However when the scale of $g(x, y, z)$ is 10^4 larger than that of $f(x, y, z)$, i.e., when the straight line is determined by Eq. 57, although we find from Fig. 11b that the relative distances converge to zero for both the parameter settings 0.3 and 30, we observe from Fig. 11a that when κ_0 is setting to be 0.3 and 30, the UAV trajectories do not converge to the desired 3D straight line. In fact we find that gaps exist between their convergent straight lines and the desired straight line.

Fig. 7 Path following results of the 2D ellipse
 $f(x, y) = x^2/250^2 + y^2/200^2 - 1 = 0$:
a UAV trajectory; **b** Relative distance; **c** Commanded course rate

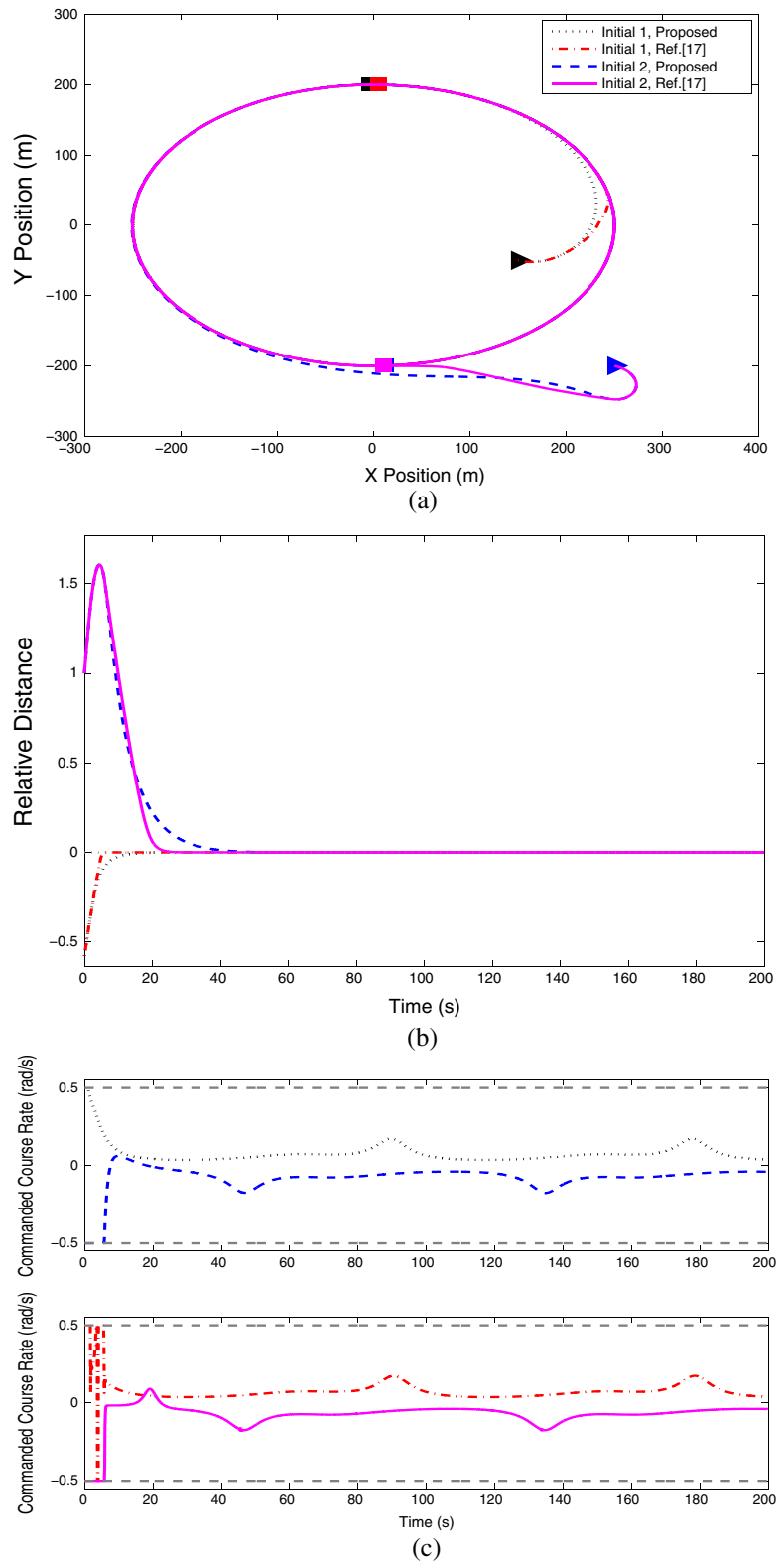


Fig. 8 Path following results of two sine curves determined by $f(x, y) = 500 \sin((x - 800)/p) + 300 - y = 0$: **a** UAV trajectory; **b** Relative distance; **c** Commanded course rate

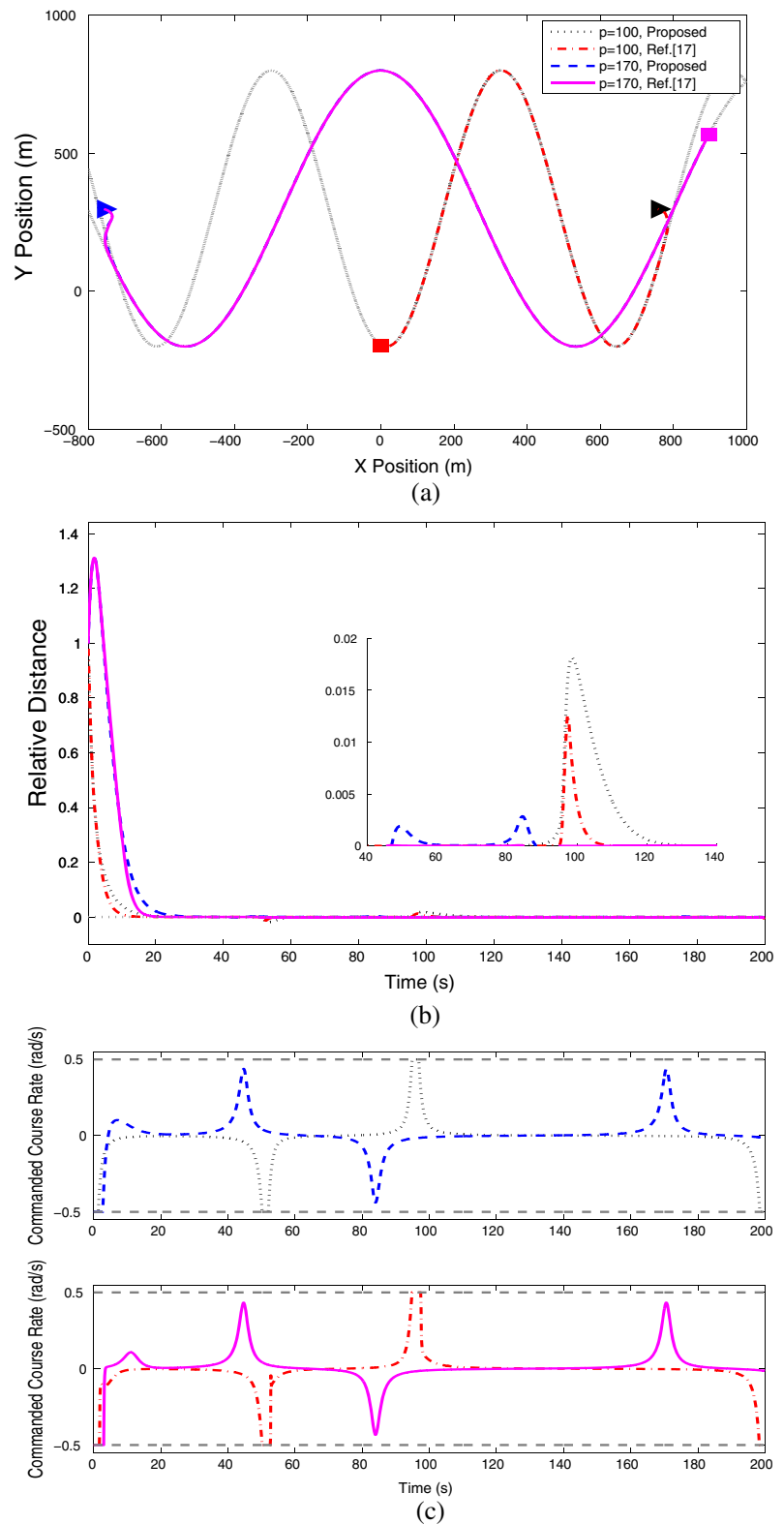


Fig. 9 Path following results of the 2D star curve determined by Eq. 55: **a** UAV trajectory; **b** Relative distance; **c** Commanded course rate

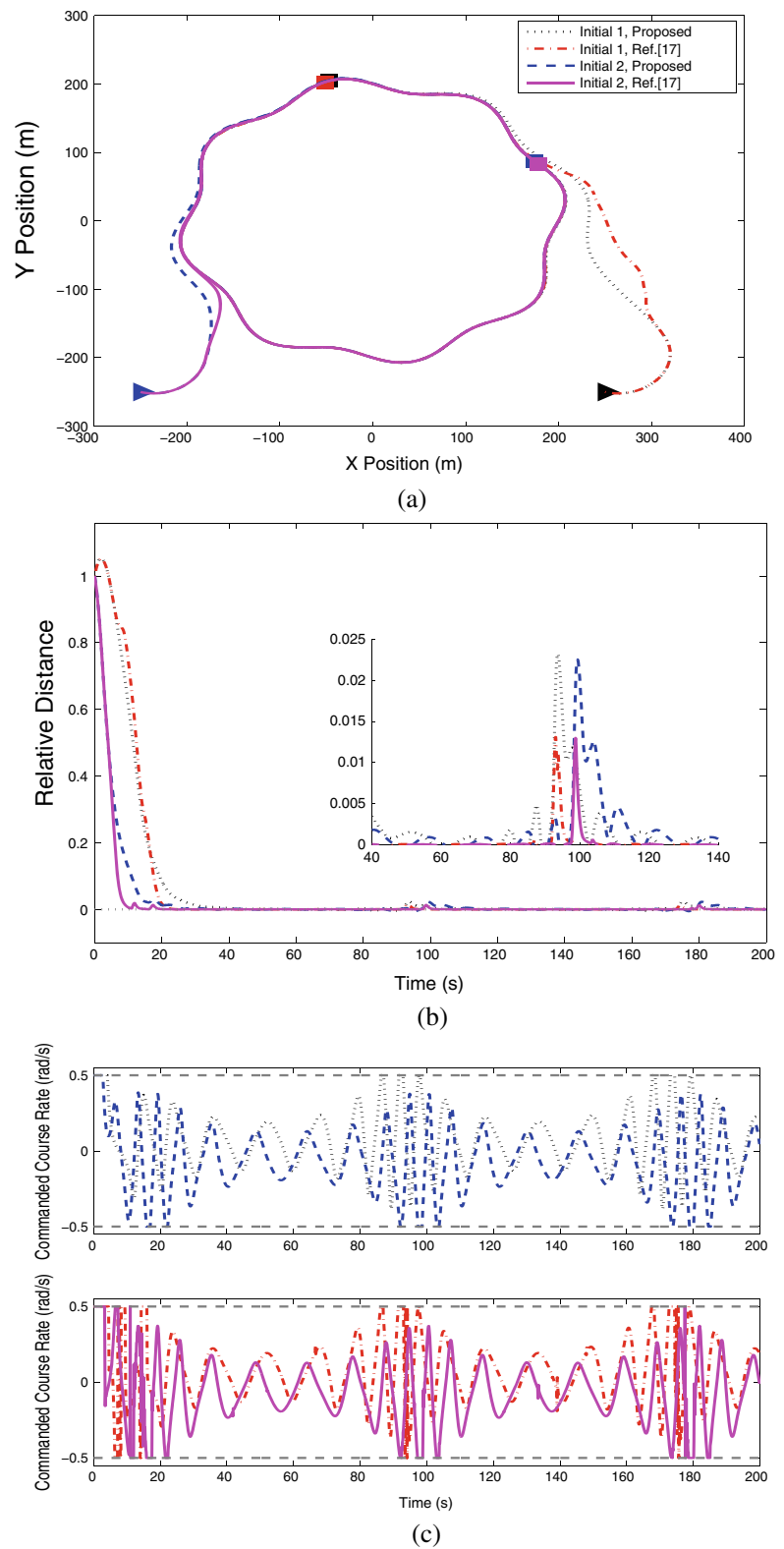


Fig. 10 Path following results of a 3D straight line when it is determined by Eq. 56: **a** UAV trajectory; **b** Relative distance; **c** Commanded course rate and climb rate

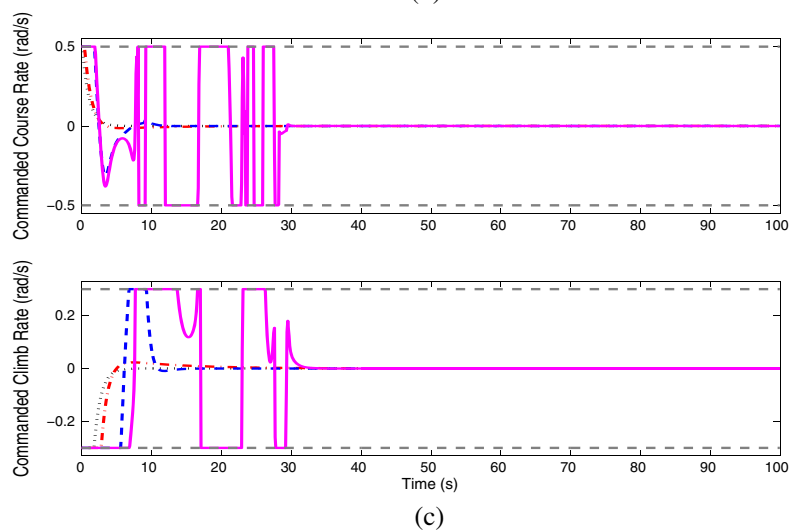
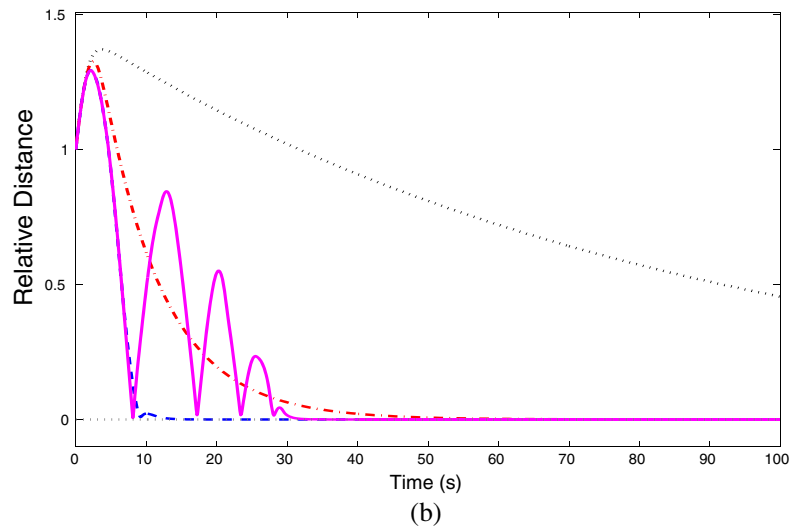
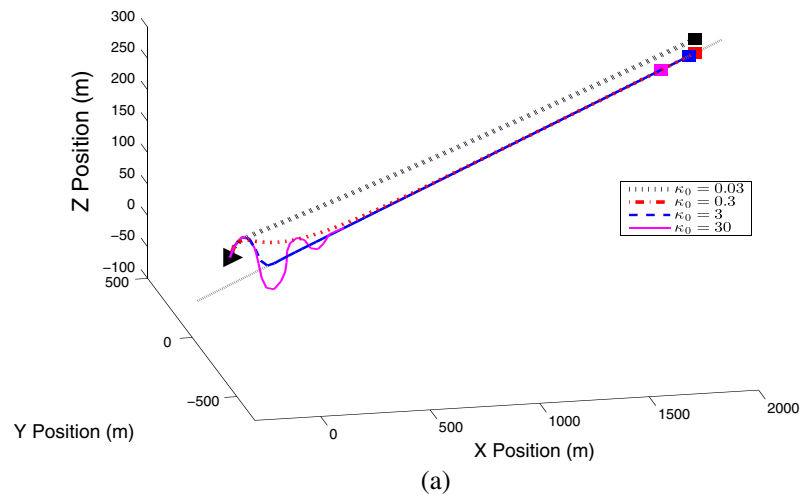


Fig. 11 Path following results of a 3D straight line when it is determined by Eq. 57: **a** UAV trajectory; **b** Relative distance; **c** Commanded course rate and climb rate

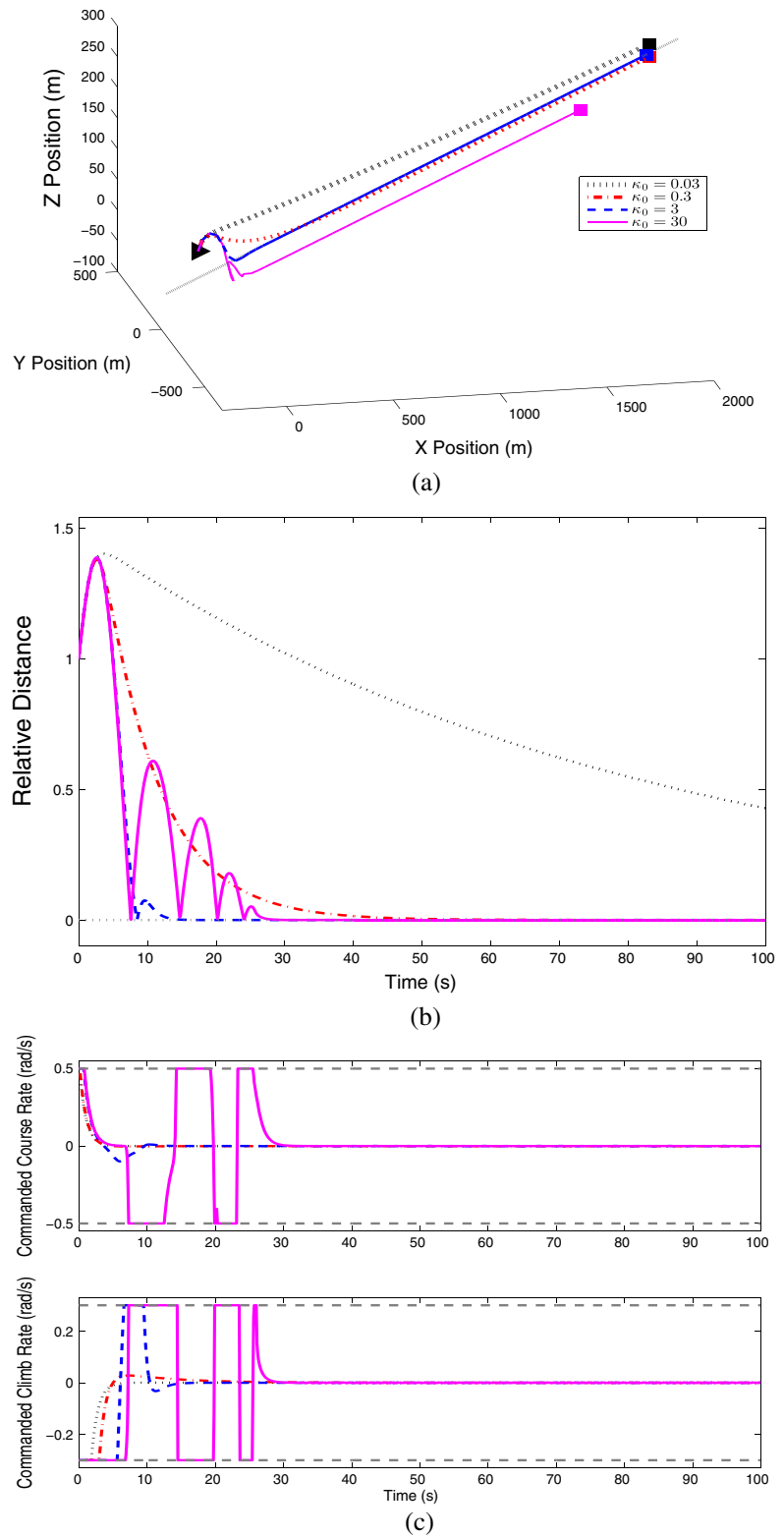


Fig. 12 Path following results of a helical curve determined by Eq. 58: **a** UAV trajectory; **b** Relative distance; **c** Commanded course rate and climb rate

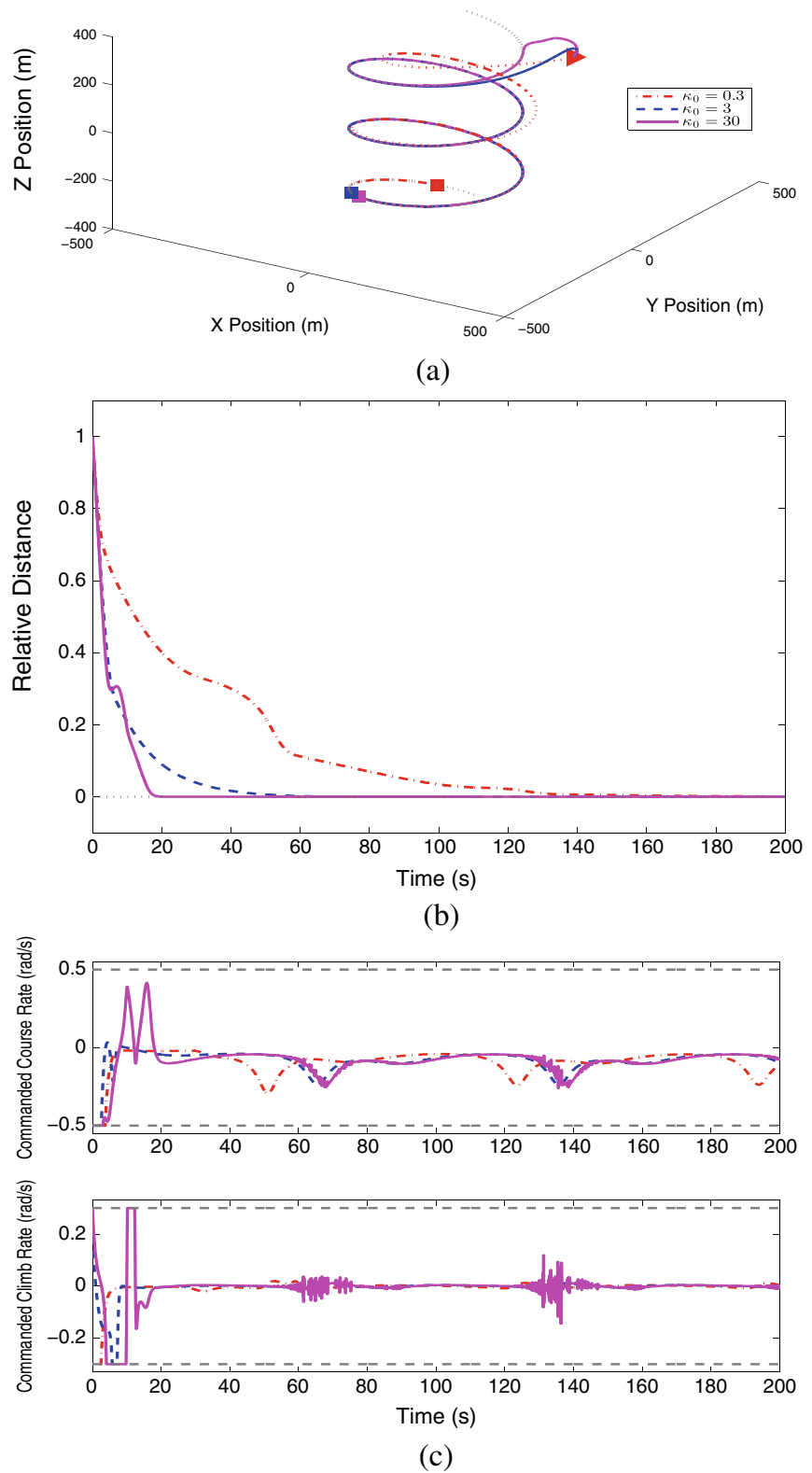
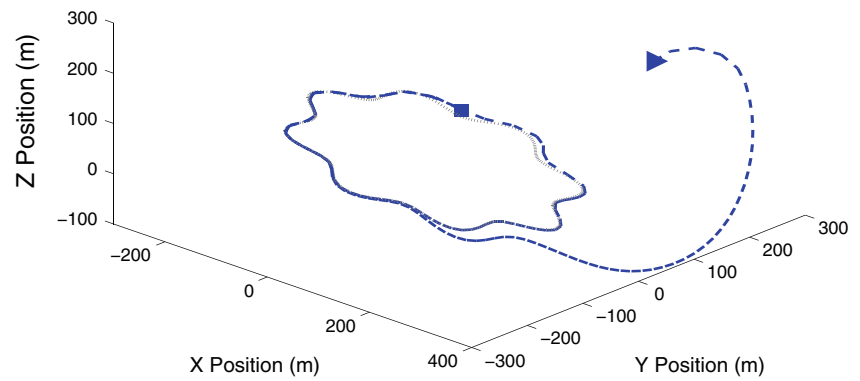
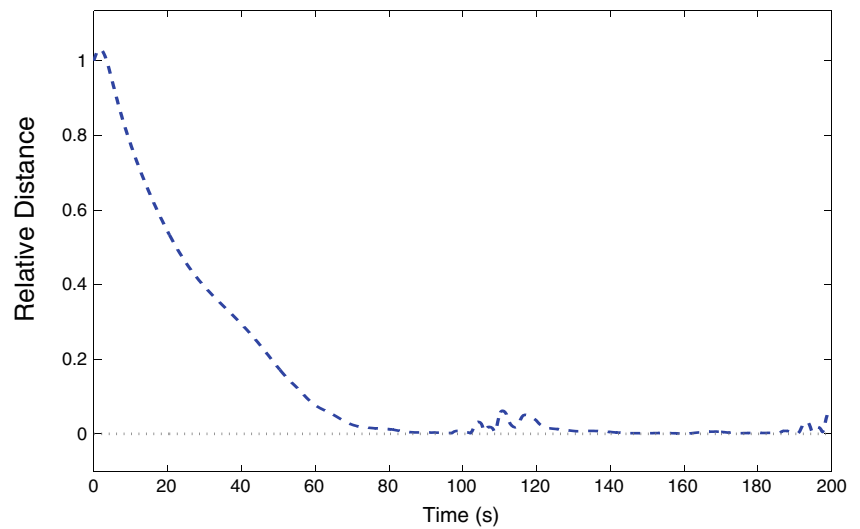


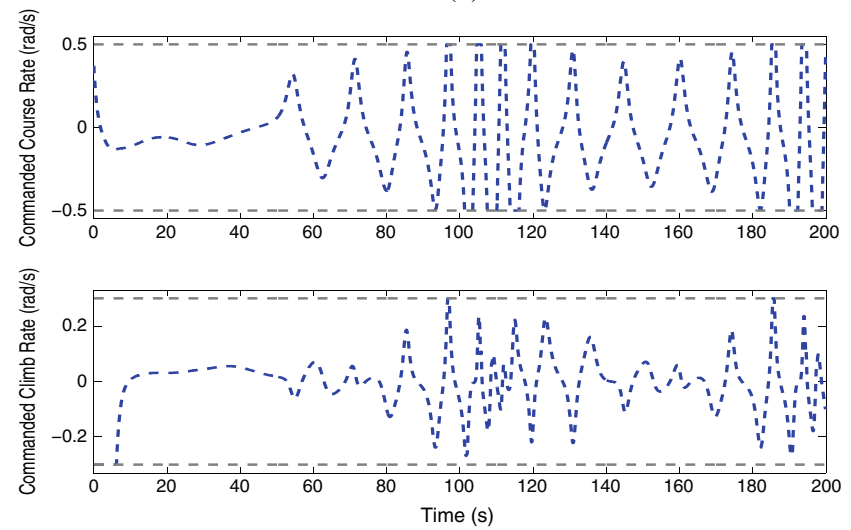
Fig. 13 Path following results of a 3D star curve determined by Eq. 59: **a** UAV trajectory; **b** Relative distance; **c** Commanded course rate and climb rate



(a)



(b)



(c)

5.4 3D Case: Performance Analysis

In this subsection, we firstly consider the path following of a helical curve determined by

$$\begin{aligned} f(x, y, z) &= x - 200 \cos(z/50) = 0, \\ g(x, y, z) &= y - 150 \sin(z/50) = 0 \end{aligned} \quad (58)$$

Let the UAVs initially locate at (233, 181, 200) with initial course 2.794 and initial climb angle 0.363. Set κ_0 as 0.3, 3 and 30. The path following results are given in Fig. 12, from which we can see that UAV trajectories converge to the desired helical curve in the end. Similarly as before when $\kappa_0 = 0.03$, the path following convergence is very slow, and thus we do not give the corresponding result.

We then consider a 3D star curve determined by

$$\begin{aligned} f(x, y, z) &= \frac{x + y + 3z}{\sqrt{5x^2 + xy + 5y^2} + 1} = 0, \\ g(x, y, z) &= \frac{1}{(5x^2 + xy + 5y^2)^{9/2} + 1} \left[- (5x^2 + xy + 5y^2)^{9/2} \right. \\ &\quad + 300\sqrt{2}(5x^2 + xy + 5y^2)^4 + 45\sqrt{22}(x^2 - y^2)(x^2 \\ &\quad + 20xy + y^2)(49x^4 - 20x^3y - 300x^2y^2 - 20xy^3 \\ &\quad \left. + 49y^4) \right] \end{aligned} \quad (59)$$

Note that the denominators $\sqrt{5x^2 + xy + 5y^2} + 1$ and $(5x^2 + xy + 5y^2)^{9/2} + 1$ are introduced to reduce the influences of the scales of the curve functions on the path following performance. The initial position of the UAV is the same with that for the helical curve path following. The initial course and climb angle are set to be 0.116 and 1.001 respectively. The alternative vector field parameter is set to be $\kappa_0 = 3$. Figure 13 shows us the 3D star curve path following result. It can be seen that except the time constants when the course rate and/or the climb rate are saturated, the UAV can well follow this 3D star curve.

6 Conclusions And Future Works

In this paper we have investigated the path following problems of 2D arbitrary twice differentiable curves and 3D arbitrary twice differentiable curves using UAVs with input constraints. 2D and 3D combined guidance vector fields based on Helmholtz theorem

have been constructed, and their globally feasible conditions are presented. Saturated flight path angle rate controllers, which conform to the input constraints of the UAVs, have been presented by tracking these vector fields. Under the globally feasible condition for the 2D combined vector field, the 2D path following error has been proven to be asymptotically convergent to zero if the saturated course rate controller is used. The 2D simulations show us that the proposed 2D curved path following approach can provide comparable performance with that proposed in Ref. [17] but with only one designing parameter and input constraints explicitly addressed. And the 3D simulations verify the effectiveness of the proposed 3D curved path following approach.

Future works include the following four aspects.

- More effective strategies to resolve the approach limitations should be further studied.
- A similar rigorous proof as that given for the 2D curved path following approach should be given for the 3D one. This is not an easy work since the course and climb angle are coupled in Eqs. 47 and 48.
- Possible extension of the proposed path following approaches to the trajectory tracking problem is to be discussed. Specifically the questions whether the proposed path following approaches can be modified to address the time-parameterized arbitrary curved paths and furthermore how to modify should be investigated.
- For the circular path following, an equiangular formation is usually considered when multiple UAVs are used to accomplish varieties of missions. Taking target tracking mission as an example, this formation ensures the coverage of the target and also the theoretical maximum amount of information about the target collected by the UAV sensors. Whether meaningful formations exist for other types of path following problems and how to use the proposed approaches to achieve a possible formation flight for multiple UAVs may be an interesting work.

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Appendix A: Proof of Theorem 2

It can be observed that if the following expression is true

$$\begin{cases} \dot{\chi}_d \geq -\dot{\chi}_{\max} + k_1 \sin \chi_e, & \text{if } \chi_e \in (0, \pi] \\ \dot{\chi}_d \leq \dot{\chi}_{\max} + k_1 \sin \chi_e, & \text{if } \chi_e \in (-\pi, 0] \end{cases} \quad (\text{A.1})$$

then we have $|\dot{\chi}_d| \leq \dot{\chi}_{\max}$ no matter what the course error χ_e is. Therefore we just need to prove that if we have Eq. 23, then Eq. A.1 satisfies.

This can be concluded from the following four cases.

Case 1. $\chi_e \in (0, \pi)$

In this case, we have $\sin \chi_e \in (0, 1)$. Consider the following function

$$\begin{aligned} F(\chi_e) &= \frac{1}{\sin \chi_e} (\dot{\chi}_d + \dot{\chi}_{\max}) \\ &= \frac{1}{\sin \chi_e} [V_g (\mathcal{A}_x \cos \chi + \mathcal{A}_y \sin \chi) + \dot{\chi}_{\max}] \\ &= V_g (\mathcal{A}_x \cos \chi_d + \mathcal{A}_y \sin \chi_d) \cot \chi_e \\ &\quad + \dot{\chi}_{\max} \csc \chi_e + V_g (-\mathcal{A}_x \sin \chi_d + \mathcal{A}_y \cos \chi_d) \end{aligned} \quad (\text{A.2})$$

Differentiating this function with respect to χ_e yields

$$\frac{dF(\chi_e)}{d\chi_e} = \frac{-V_g (\mathcal{A}_x \cos \chi_d + \mathcal{A}_y \sin \chi_d) - \dot{\chi}_{\max} \cos \chi_e}{\sin^2 \chi_e} \quad (\text{A.3})$$

Letting this derivative be zero, we have

$$\cos \chi_{e0} = -\frac{V_g (\mathcal{A}_x \cos \chi_d + \mathcal{A}_y \sin \chi_d)}{\dot{\chi}_{\max}} \quad (\text{A.4})$$

From Eq. 23 we get

$$|\cos \chi_{e0}| \leq \frac{V_g \sqrt{\mathcal{A}_x^2 + \mathcal{A}_y^2}}{\dot{\chi}_{\max}} \leq \frac{7}{10} \quad (\text{A.5})$$

$$\sin \chi_{e0} = \sqrt{1 - \cos^2 \chi_{e0}} \geq \sqrt{51}/10 \quad (\text{A.6})$$

Computing the second derivative of $F(\chi_e)$ and substituting (A.4) and (A.6) into it, we obtain

$$\left. \frac{d^2 F(\chi_e)}{d\chi_e^2} \right|_{\chi_e = \chi_{e0}} = \frac{\dot{\chi}_{\max}}{\sin \chi_{e0}} > 0 \quad (\text{A.7})$$

from which we can see that Eqs. A.4 and A.6 determine a lower bound for $F(\chi_e)$. This lower bound is

$$\begin{aligned} F_{\min}(\chi_e) &= V_g (-\mathcal{A}_x \sin \chi_d + \mathcal{A}_y \cos \chi_d) + \dot{\chi}_{\max} \sin \chi_{e0} \\ &\geq -V_g \sqrt{\mathcal{A}_x^2 + \mathcal{A}_y^2} + \dot{\chi}_{\max} \sin \chi_{e0} \\ &\geq \frac{\sqrt{51}-7}{10} \dot{\chi}_{\max} > 0 \end{aligned} \quad (\text{A.8})$$

By letting $0 < k_1 < \frac{\sqrt{51}-7}{10} \dot{\chi}_{\max}$, we get that $F(\chi_e) \geq k_1$. From the definition of $F(\chi_e)$ in Eq. A.2, we conclude that $\dot{\chi}_d \geq -\dot{\chi}_{\max} + k_1 \sin \chi_e$.

Case 2. $\chi_e \in (-\pi, 0)$

In this case, $\sin \chi_e \in (-1, 0)$. By considering another function

$$F(\chi_e) = \frac{1}{\sin \chi_e} (\dot{\chi}_d - \dot{\chi}_{\max}) \quad (\text{A.9})$$

the conclusion $\dot{\chi}_d \leq \dot{\chi}_{\max} + k_1 \sin \chi_e$ can be drawn similarly as before by taking the same k_1 .

Case 3. $\chi_e = 0$

In this case, we get

$$\begin{aligned} \dot{\chi}_d &= V_g (\mathcal{A}_x \cos \chi_d + \mathcal{A}_y \sin \chi_d) \\ &\leq V_g \sqrt{\mathcal{A}_x^2 + \mathcal{A}_y^2} \leq \frac{7\dot{\chi}_{\max}}{10} < \dot{\chi}_{\max} \end{aligned} \quad (\text{A.10})$$

Then by choosing an arbitrary positive k_1 , it can be easily seen that Eq. A.1 holds since in this case $\sin \chi_e = 0$.

Case 4. $\chi_e = \pi$

In this case

$$\begin{aligned} \dot{\chi}_d &= -V_g (\mathcal{A}_x \cos \chi_d + \mathcal{A}_y \sin \chi_d) \\ &\geq -V_g \sqrt{\mathcal{A}_x^2 + \mathcal{A}_y^2} \geq -\frac{7\dot{\chi}_{\max}}{10} > -\dot{\chi}_{\max} \end{aligned} \quad (\text{A.11})$$

Similarly an arbitrary positive k_1 can be chosen to make Eq. A.1 hold since $\sin \pi = 0$.

Appendix B: Proof of Lemma 1

From Eqs. 20–21 we have

$$\dot{\chi}_e = \dot{\chi} - \dot{\chi}_d = \begin{cases} -k_\chi \chi_e, & \text{if } |-k_\chi \chi_e + \dot{\chi}_d| \leq \dot{\chi}_{\max} \\ \dot{\chi}_{\max} - \dot{\chi}_d, & \text{if } -k_\chi \chi_e + \dot{\chi}_d > \dot{\chi}_{\max} \\ -\dot{\chi}_{\max} - \dot{\chi}_d, & \text{if } -k_\chi \chi_e + \dot{\chi}_d < -\dot{\chi}_{\max} \end{cases} \quad (\text{A.12})$$

Let us discuss this from two cases.

Case 1. $\chi_e \in (0, \pi]$

In this case, $\sin \chi_e \geq 0$. If $|-k_\chi \chi_e + \dot{\chi}_d| \leq \dot{\chi}_{\max}$, then $\dot{\chi}_e = -k_\chi \chi_e$. Then we have $\dot{\chi}_e \leq -k_2 \sin \chi_e$ by choosing a k_2 such that $0 < k_2 \leq k_\chi$. If $-k_\chi \chi_e + \dot{\chi}_d > \dot{\chi}_{\max}$, then $\dot{\chi}_e = \dot{\chi}_{\max} - \dot{\chi}_d < -k_\chi \chi_e$. The same k_2 can be chosen to make $\dot{\chi}_e \leq -k_2 \sin \chi_e$. Finally if $-k_\chi \chi_e + \dot{\chi}_d < -\dot{\chi}_{\max}$, we can see from Eq. A.1 that $\dot{\chi}_e = -\dot{\chi}_{\max} - \dot{\chi}_d \leq -k_1 \sin \chi_e$. Then a k_2 satisfying $0 < k_2 \leq k_1$ can be chosen to make $\dot{\chi}_e \leq -k_2 \sin \chi_e$. In conclusion we obtain that there exists a positive constant k_2 subject to $k_2 \leq \min\{k_\chi, k_1\}$ that can ensure $\dot{\chi}_e \leq -k_2 \sin \chi_e$.

Case 2. $\chi_e \in (-\pi, 0]$

Following similar analysis as given in Case 1, it can be easily concluded that the same positive constant $0 < k_2 \leq \min\{k_\chi, k_1\}$ can be chosen to make that $\dot{\chi}_e \geq -k_2 \sin \chi_e$.

Appendix C: Proof of Lemma 2

It can be seen from Lemma 1 that $\dot{\chi}_e / \sin \chi_e \leq -k_2$. Denoting the initial course error as χ_{e0} , and noticing that

$$\frac{\dot{\chi}_e}{\sin \chi_e} = \frac{d[\log |\tan(\frac{\chi_e}{2})|]}{dt} \quad (\text{A.13})$$

we obtain

$$\left| \tan\left(\frac{\chi_e}{2}\right) \right| \leq \left| \tan\left(\frac{\chi_{e0}}{2}\right) \right| e^{-k_2 t} \quad (\text{A.14})$$

Note that the following is based on the combined vector field parameter choice $s = +1$, the situation when $s = -1$ is similar.

Evaluating the time derivative of $r = f(x, y)$ gives us

$$\begin{aligned} \dot{r} &= f_x \dot{x} + f_y \dot{y} = V_g \sqrt{f_x^2 + f_y^2} \sin(\chi - \chi_p) \\ &= V_g \sqrt{f_x^2 + f_y^2} \sin(\chi_e + \chi_r) \end{aligned} \quad (\text{A.15})$$

where $\chi_p = \arctan2(-f_x, f_y)$, and $\chi_r = \chi_d - \chi_p$. It needs to be noted that χ_p is the tangential direction of the curve corresponding to the current UAV location. Simplification of $\cos(\chi_d - \chi_p)$ yields that $\cos \chi_r = \text{sech}(\kappa f) \geq 0$, from which we see that $\chi_r \in (-\pi/2, \pi/2)$. Similarly we have $\sin \chi_r = -\tanh(\kappa f)$, from which we further find that $\chi_r \in (-\pi/2, 0]$ when $r \geq 0$ and $\chi_r \in [0, \pi/2)$ when $r \leq 0$.

The conclusion are proved from the following two cases.

Case 1. $r \geq 0$

When $\chi_e \in (0, \pi]$, according to Lemma 1, we have $\dot{\chi}_e \leq -k_2 \sin \chi_e \leq 0$, which indicates that $0 \leq \chi_e \leq \chi_{e0} \leq \pi$. This further leads to $-\pi/2 \leq \chi_e + \chi_r \leq \chi_{e0} + \chi_r \leq \pi$. If $-\pi/2 \leq \chi_{e0} + \chi_r \leq 0$, then we find that $\dot{r} \leq 0$ from Eq. A.15. This results in $r \leq r_0$. If $0 < \chi_{e0} + \chi_r \leq \pi$, then $\dot{r} \geq 0$, which implies that r will increase. Since the course error χ_e and χ_r as well are decreasing, the increasing of r stops when $\chi_e + \chi_r$ becomes 0, i.e., when $\chi_e = -\chi_r > 0$. According to Eq. A.14, an upper bound of the time period for χ_e changing from χ_{e0} to $-\chi_{r0}$ can be given by

$$t_1 = \log \left| \frac{\tan(\frac{\chi_{e0}}{2})}{\tan(\frac{-\chi_{r0}}{2})} \right| \quad (\text{A.16})$$

Note that instead of the unknown eventual χ_r , we use the conservative initial counterpart χ_{r0} . Then an upper bound for r can be computed as $r \leq r_1 = r_0 + M(V_a + W)t_1$. Here $\dot{r} \leq V_g M \leq (V_a + W)M$ is used.

When $\chi_e \in (-\pi, 0]$, Eq. 26 gives us that $\dot{\chi}_e \geq -k_2 \sin \chi_e \geq 0$. This further gives $-\pi \leq \chi_{e0} \leq \chi_e \leq 0$. Recalling that $\chi_r \in (-\pi/2, 0]$, we have $-3\pi/2 \leq \chi_{e0} + \chi_r \leq \chi_e + \chi_r \leq 0$. If $-\pi \leq \chi_{e0} + \chi_r \leq 0$, then from Eq. A.15 we can see that $\dot{r} \leq 0$, which means $r \leq r_0$. If $-3\pi/2 \leq \chi_{e0} + \chi_r \leq -\pi$, then $\dot{r} \geq 0$. Similarly r increases until $\chi_e + \chi_r = -\pi$, i.e., $\chi_e = -\pi - \chi_r < 0$. The time period of r 's increasing is not more than

$$t_2 = \log \left| \frac{\tan(\frac{\chi_{e0}}{2})}{\tan(\frac{-\pi - \chi_{r0}}{2})} \right| \quad (\text{A.17})$$

In this case the upper bound for r is $r_2 = r_0 + M(V_a + W)t_2$.

It can be seen from the above discussion that $0 \leq r \leq K_1 = \max\{r_1, r_2\}$, which means that $|r|$ is upper bounded.

Case 2. $r < 0$

Following similar analysis for the case when $r < 0$, we find that there exists a positive constant such that $|r| \leq K_2$.

Then the positive constant K which is used to bound r can be chosen as $K = \max\{K_1, K_2\}$.

Appendix D: Proof of Lemma 3

Denote $f(x) = x \tanh(\kappa x) \geq 0$ and $g(x) = \rho x^2 \geq 0$. Then we can get that

$$\begin{aligned} \frac{df(x)}{dx} &= \tanh(\kappa x) + \kappa x \operatorname{sech}^2(\kappa x) \geq \kappa x \operatorname{sech}^2(\kappa x) \\ &\geq \kappa x \operatorname{sech}^2(\kappa K) \geq 2\rho x = \frac{dg(x)}{dx} \end{aligned} \quad (\text{A.18})$$

In combination with the fact $f(0) = g(0) = 0$, we can see that if $0 < \rho \leq \kappa \operatorname{sech}^2(\kappa K)/2$, then we have $x \tanh(\kappa x) \geq \rho x^2$ for $0 \leq x \leq K$.

Appendix E: Proof of Theorem 5

We prove the satisfactions of Eqs. 51a and 51b respectively as follows.

(i) Due to the same reason of the introduction of Eq. A.1, to prove $|\dot{\gamma}_d| \leq \dot{\gamma}_{\max}$, we just need to prove

$$\begin{cases} \dot{\gamma}_d \geq -\dot{\gamma}_{\max} + k_1 \sin \gamma_e, & \text{if } \gamma_e \in (0, \pi] \\ \dot{\gamma}_d \leq \dot{\gamma}_{\max} + k_1 \sin \gamma_e, & \text{if } \gamma_e \in (-\pi, 0] \end{cases} \quad (\text{A.19})$$

This can be concluded from the following four cases.

Case 1. $\gamma_e \in (0, \pi)$

In this case, we have $\sin \gamma_e \in (0, 1)$. Consider the following function

$$\begin{aligned} F(\gamma_e) &= \frac{1}{\sin \gamma_e} (\dot{\gamma}_d + \dot{\gamma}_{\max}) \\ &= \frac{1}{\sin \gamma_e} [V_g ((C_x \cos \chi + C_y \sin \chi) \cos \gamma \\ &\quad + C_z \sin \gamma) + \dot{\gamma}_{\max}] \\ &= V_g [(C_x \cos \chi + C_y \sin \chi) \cos \gamma_d + C_z \sin \gamma_d] \cot \gamma_e \\ &\quad + \dot{\gamma}_{\max} \csc \gamma_e + V_g [-(C_x \cos \chi + C_y \sin \chi) \sin \gamma_d \\ &\quad + C_z \cos \gamma_d] \end{aligned} \quad (\text{A.20})$$

Following similar procedure as in the proof of Theorem 2, we find that when

$$\cos \gamma_{e0} = -\frac{V_g [(C_x \cos \chi + C_y \sin \chi) \cos \gamma_d + C_z \sin \gamma_d]}{\dot{\gamma}_{\max}} \leq \frac{7}{10} \quad (\text{A.21})$$

$F(\gamma_e)$ is minimized, and its minimization value is

$$\begin{aligned} F_{\min}(\gamma_e) &= V_g [-(C_x \cos \chi + C_y \sin \chi) \sin \gamma_d \\ &\quad + C_z \cos \gamma_d] + \dot{\gamma}_{\max} \sin \gamma_{e0} \\ &\geq -V_g \sqrt{C_x^2 + C_y^2 + C_z^2} + \dot{\gamma}_{\max} \sin \gamma_{e0} \\ &\geq \frac{\sqrt{51}-7}{10} \dot{\gamma}_{\max} > 0 \end{aligned} \quad (\text{A.22})$$

By letting $0 < k_1 < \frac{\sqrt{51}-7}{10} \dot{\gamma}_{\max}$, we get that $F(\gamma_e) \geq k_1$. From the definition of $F(\gamma_e)$ in Eq. A.20, we conclude that $\dot{\gamma}_d \geq -\dot{\gamma}_{\max} + k_1 \sin \gamma_e$.

Case 2. $\gamma_e \in (-\pi, 0)$

In this case, $\sin \gamma_e \in (-1, 0)$. By considering another function

$$F(\gamma_e) = \frac{1}{\sin \gamma_e} (\dot{\gamma}_d - \dot{\gamma}_{\max}) \quad (\text{A.23})$$

the conclusion $\dot{\gamma}_d \leq \dot{\gamma}_{\max} + k_1 \sin \gamma_e$ can be drawn similarly in **Case 1** by taking the same k_1 .

Case 3. $\gamma_e = 0$

In this case, we get

$$\begin{aligned} \dot{\gamma}_d &= V_g [(C_x \cos \chi + C_y \sin \chi) \cos \gamma_d + C_z \sin \gamma_d] \\ &\leq V_g \sqrt{C_x^2 + C_y^2 + C_z^2} \leq \frac{7\dot{\gamma}_{\max}}{10} < \dot{\gamma}_{\max} \end{aligned} \quad (\text{A.24})$$

Then by choosing an arbitrary positive k_1 , it can be easily seen that Eq. 51a holds since in this case $\sin \gamma_e = 0$.

Case 4. $\gamma_e = \pi$

In this case

$$\begin{aligned}\dot{\gamma}_d &= -V_g [(\mathcal{C}_x \cos \chi + \mathcal{C}_y \sin \chi) \cos \gamma_d + \mathcal{C}_z \sin \gamma_d] \\ &\geq -V_g \sqrt{\mathcal{C}_x^2 + \mathcal{C}_y^2 + \mathcal{C}_z^2} \geq -\frac{7\dot{\chi}_{\max}}{10} > -\dot{\chi}_{\max}\end{aligned}\quad (\text{A.25})$$

Similarly an arbitrary positive k_1 can be chosen to make Eq. 51a hold since $\sin \pi = 0$.

(ii) Similarly, to prove $|\dot{\chi}_d| \leq \dot{\chi}_{\max}$, we just need to prove (A.1). Let us consider this from the following four cases.

Case 1. $\chi_e \in (0, \pi)$

In this case, $\sin \chi_e \in (0, 1)$. Consider the function

$$\begin{aligned}F(\chi_e) &= \frac{1}{\sin \chi_e} (\dot{\chi}_d + \dot{\chi}_{\max}) \\ &= \frac{1}{\sin \chi_e} [V_g (\mathcal{B}_x \cos \gamma \cos \chi + \mathcal{B}_y \cos \gamma \sin \chi \\ &\quad + \mathcal{B}_z \sin \gamma) + \dot{\chi}_{\max}] \\ &= V_g \cos \gamma (\mathcal{B}_x \cos \chi_d + \mathcal{B}_y \sin \chi_d) \cot \chi_e \\ &\quad + (V_g \mathcal{B}_z \sin \gamma + \dot{\chi}_{\max}) \csc \chi_e \\ &\quad + V_g \cos \gamma (-\mathcal{B}_x \sin \chi_d + \mathcal{B}_y \cos \chi_d)\end{aligned}\quad (\text{A.26})$$

According to similar procedure as before, we find that when $|\mathcal{B}_z| \leq \alpha \dot{\chi}_{\max} / V_g$ and

$$\begin{aligned}\cos \chi_{e0} &= -\frac{V_g \cos \gamma (\mathcal{B}_x \cos \chi_d + \mathcal{B}_y \sin \chi_d)}{V_g \mathcal{B}_z \sin \gamma + \dot{\chi}_{\max}} \\ &\leq \frac{7(1 - \alpha)}{10}\end{aligned}\quad (\text{A.27})$$

$F(\chi_e)$ has a minimum value

$$\begin{aligned}F_{\min}(\chi_e) &= V_g \cos \gamma (-\mathcal{B}_x \sin \chi_d + \mathcal{B}_y \cos \chi_d) \\ &\quad + (V_g \mathcal{B}_z \sin \gamma + \dot{\chi}_{\max}) \sin \chi_{e0} \\ &\geq -V_g \sqrt{\mathcal{B}_x^2 + \mathcal{B}_y^2} + (-V_g |\mathcal{B}_z| + \dot{\chi}_{\max}) \sin \chi_{e0} \\ &\geq \frac{\sqrt{51} - 7}{10} (1 - \alpha) \dot{\chi}_{\max} > 0\end{aligned}\quad (\text{A.28})$$

Setting $0 < k_1 < \frac{\sqrt{51} - 7}{10} (1 - \alpha) \dot{\chi}_{\max}$, we get $F(\chi_e) \geq k_1$. Then from the definition of $F(\chi_e)$ in Eq. A.26, we conclude that $\dot{\chi}_d \geq -\dot{\chi}_{\max} + k_1 \sin \chi_e$.

Case 2. $\chi_e \in (-\pi, 0)$

In this case, $\sin \chi_e \in (-1, 0)$. Similarly, by considering the function (A.9), the conclusion $\dot{\chi}_d \leq \dot{\chi}_{\max} +$

$k_1 \sin \chi_e$ can be drawn by taking the same k_1 as that given in Case 1.

Case 3. $\chi_e = 0$

In this case

$$\begin{aligned}\dot{\chi}_d &= V_g [(\mathcal{B}_x \cos \chi_d + \mathcal{B}_y \sin \chi_d) \cos \gamma + \mathcal{B}_z \sin \gamma] \\ &\leq V_g \sqrt{\mathcal{B}_x^2 + \mathcal{B}_y^2} + V_g |\mathcal{B}_z| \leq \frac{(7 + 3\alpha) \dot{\chi}_{\max}}{10} < \dot{\chi}_{\max}\end{aligned}\quad (\text{A.29})$$

Then by choosing an arbitrary positive k_1 , it can be easily observed that Eq. A.1 holds since in this case $\sin \chi_e = 0$.

Case 4. $\chi_e = \pi$

In this case, we have

$$\begin{aligned}\dot{\chi}_d &= -V_g [(\mathcal{B}_x \cos \chi_d + \mathcal{B}_y \sin \chi_d) \cos \gamma + \mathcal{B}_z \sin \gamma] \\ &\geq -V_g \sqrt{\mathcal{B}_x^2 + \mathcal{B}_y^2} - V_g |\mathcal{B}_z| \geq -\frac{(7 + 3\alpha) \dot{\chi}_{\max}}{10} \\ &> -\dot{\chi}_{\max}\end{aligned}\quad (\text{A.30})$$

Similarly an arbitrary positive k_1 can be chosen to guarantee (A.1)'s satisfaction since in this case $\sin \chi_e = 0$.

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Yueqian Liang was born in Shandong, China. He received his B.S. degree in Information and Computation Science from Minzu University of China, Beijing, China, in 2008. He is now working towards his Ph.D. degree in Control Theory and Control Engineering at Beihang University (BUAA), Beijing, China. His current research interests include target tracking using unmanned aerial vehicles.

Yingmin Jia received his B.S. in Control Theory from Shandong University, Ji’nan, China, in January 1982, and his M.S. and Ph.D. both in Control Theory and Applications from Beihang University (BUAA), Beijing, China, in 1990 and 1993, respectively. In 1993, he joined the Seventh Research Division at Beihang University, where he is currently Professor of automatic control. From February 1995 until February 1996, he was a visiting professor with the Institute of Robotics and Mechatronics of the German Aerospace Center (DLR), Oberpfaffenhofen, Germany. He held an Alexander von Humboldt (AvH) research fellowship with the Institute of Control Engineering at the Technical University Hamburg-Harburg, Hamburg, Germany, from December 1996 until March 1998, and a JSPS research fellowship with the Department of Electrical and Electronic Systems at the Osaka Prefecture University, Osaka, Japan, from March 2000 until March 2002. He was a visiting professor with the Department of Statistics at the University of California Berkeley from December 2006 until March 2007. His current research interests include robust control, adaptive control and intelligent control, and their applications in industrial processes and vehicle systems. He is author and co-author of numerous papers and of the book ‘Robust H_∞ Control’ (Science Press 2007).